

Financial Constraints and Trade Credit as a Strategic Tool: Evidence from Small-Scale Reservation Reforms in India

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Abstract

How financial constraints affect incumbent supplier firm's choice of extending more trade credit vis-à-vis offering price discounts when facing increased threat of entry by competitors ? Threatened incumbent supplier firms may extend more trade credit, *ex-ante*, to defend their market power, or they may reduce prices or both. I test these predictions by exploiting plausibly exogenous, staggered removals of product level entry barriers for Indian manufacturing firms, and find that an average incumbent supplier firm extends 11% more trade credit and lowers price by 8.6% with an increased threat of entry. Interestingly, firms with deeper pockets offer longer terms on trade credit, while constrained firms rely on price discounts. Also, I find that financially constrained customers take credit offers while unconstrained customers take price discounts. I further confirm my results using another policy change in the country that improved access to finance. Thus emphasize the role of trade credit as a strategic tool to defend market power, when firms face financial constraints.

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Introduction

Inter-firm credit, also known as trade credit, finances around 90% of the world merchandise trade amounting to \$14 trillion (William (2008)). In the United States, for example, 80% of manufacturing firms offer their product on trade credit.¹ Recent empirical works have focused on understanding the role of trade credit supply in corporate bankruptcies, probability of trade and its effect on industry structure.² However, the effect of industry structure on trade credit supply – especially its role in the context of defending product market power – has received little attention in the literature.

Whenever firms face entry threat by new competitors, incumbent firms undertake various preemptive actions to deter entry.³ Further, a large literature predicts that “deep-pocketed firms” will attempt to drive financially constrained competitors out of business (see e.g. Telser (1966), Bolton and Scharfstein (1990)). One imperative question in this context concerns whether financial constraints affects firms choice of strategic action i.e. extending more trade credit vis-à-vis offering price discounts, *ex-ante* when faced with a credible threat of increased competition. The theory presents the following hypothesis here: incumbent supplier firms may choose to offer longer payment terms or price discounts or both. But the choice may vary with the financial strength of both supplier and customer firms. On the one hand, incumbent firms with deep pockets may choose to extend trade credit at longer payment terms to defend their market share when customers are financially constrained i.e. when customer value a dollar of credit more than a dollar. The longer credit terms increase barriers to entry for potential competitors with shallow pockets. Also, information asymmetry related to customer’s

¹ See Tirole (2010).

² Jacobson and von Schedvin (2015), using Swedish data, find that trade debtor (customers) failures are associated with substantially enhanced bankruptcy risks for the trade creditors (suppliers). Breza and Liberman (2016), using large retailer data, find that the restriction reduces the likelihood of trade by 11%. Barrot (2016) finds, in French trucking industry, long payment terms impose substantial *liquidity risk* on financially weaker suppliers, and force them into financial distress more often than they would if they were paid earlier.

³ Limit-pricing (Salop (1979) and Milgrom and Roberts (1982)), capacity building/investment (Dixit (1980), Goolsbee and Syverson (2008), Frésard and Valta (2015)) and leverage (Brander and Lewis (1986), Cookson (2016)).

creditworthiness further dampens the potential entrant's ability to match the same strategy. On the other hand, the incumbent supplier firm may choose to offer price discounts to its customers when supplying firm is credit constrained and customer firm is not.

However, empirically it is challenging to test the above hypothesis in the data. The primary difficulty lies in cleanly identifying firms' choice of *ex-ante* strategic actions in response to a *threat of entry* as distinct from actual entry. First, theoretically, firms' trade credit policies, product pricing, and other product market strategies are jointly determined in equilibrium. Therefore, clean identification makes it essential to examine situations where firms receive a *exogenous shock* in the product market. Second, the shock has to ideally affect the *threat of entry*, as opposed to *actual entry* by a competing firm. It is important because *actual entry* by a competing firm is already an outcome of the trade credit strategies or price policies adopted by incumbent supplier firms. Moreover, the purpose of changing corporate policies such as trade credit policy or product prices after a rival has already entered could be very different from the incentives to change policy before entry. For example, whether or not trade credit can act as a pre-emptive tool to deter entry cannot be examined if the rival firm has already entered, or is already producing the competing product (in the case of tariff cuts). Finally, to make a comparison between the alternative strategies empirically, it's challenging to find data on both trade credit and product prices simultaneously.

I attempt to solve the identification and data challenge by exploiting the staggered removal of product-level entry barriers for Indian manufacturing firms as *plausibly* exogenous shocks to the threat of entry, which are unaffected by existing firm policies of incumbent supplier firms. I find that an average incumbent supplier firm extends 11% more trade credit and lowers price by 8.6% with an increased threat of entry. Also, I find that "deep pocketed" firms offer more credit to their customers, while financially constrained customers to take more credit from their suppliers, consistent with the above hypothesis.

The institutional setting I examine concerns the removal of manufacturing restrictions for

a list of products by the Government of India (GoI), by which the right to manufacture these products was reserved for small-scale or export oriented large firms.⁴ Following India's 1991 trade liberalization, the GoI appointed a special committee to reconsider the list of reserved items in 1995 (MSME (2007)). Based on recommendations from this committee, this reservation policy was dismantled in a *staggered* manner for about 1,024 products starting in 1997 (also known as dereservation), and resulted in the near complete removal of reservations by 2008 (Figure 1).⁵ For example, 15 products were dereserved in 1997 which included ice-cream, poultry feed, hair dryers, ashtray, etc.

The use of this particular institutional setting confers the following advantages. Firstly, the product dereservation was arbitrary in nature. A reading of government documents, reports, and media does not give clear-cut reasons as to why certain products were initially reserved or dereserved at certain times, and even policy-makers in India at this time recognize the arbitrariness of the product list subject to dereservation in various years (Hussain (1997), Tewari and Wilde (2014) and Martin, Nataraj, and Harrison (2016)). In my analysis, I estimate a hazard model for timing of dereservation, and do not find evidence for existing trade credit policies of incumbent supplier firms predicting the event timing.⁶ Secondly, the removal of entry barriers does seem to have had real effects regarding increased competition which took effect in a slow manner.⁷ I find in a difference-in-difference setting that market concentration index (HHI) declined significantly within few years of dereservation in the affected industries. This is consistent with the notion that it takes time for new entrants to install plant or expand capacity to manufacture dereserved products. Consistent with economic theory, I also find

⁴The identification strategy is similar to Martin, Nataraj, and Harrison (2016), who find positive impact on employment growth after the removal of entry barrier.

⁵ Appendix A, Table A1 shows sample notification about list of items dereserved in 1997, while Table A2 provides information on timing of dereservation by industry-year.

⁶ In a robustness check, in treatment group, I keep products that are not produced by large leading firms in the country and find similar results.

⁷ Survey of small firms by Federation of Indian Chambers of Commerce and Industry (FICCI) indicates that small-scale firms see entry of multinational corporations (MNCs) and dereservation of the items reserved for them as a threat to their market share (http://www.business-standard.com/article/specials/ssis-feel-threatened-by-mncs-dereservation-ficci-survey-197122901002_1.html).

entry rate going up and price index going down for dereserved products. This unique setting allows me to distinguish firms' ex-ante strategies when facing the threat of entry. I specifically focus on a short window around dereservation to identify firms' pre-emptive action to increased *entry threat*.

Using this unique regulation change affecting Indian manufacturing firms, I implement a difference-in-differences (DID) approach to estimate the effect of dereservation on the provision of trade credit and product prices. Specifically, I examine whether *incumbent* firms (those producing the reserved product before the reform), *ex-ante*, extend more credit to their customers when they face increased entry threat by new competitors (those that may move into the product space after the product was de-reserved). I find that an average incumbent supplier firm extends around 11% more trade credit (measured as the ratio of accounts receivable/sales) surrounding removal of entry barriers, as compared to their counterparts who do not face any such change in entry barriers. For an average incumbent supplier firm with trade credit payment terms of about two months, this translates to an increase in payment terms by around 8-10 days. Also, in the pre-dereservation period, I find no differences in the ratio of accounts receivable/sales (a proxy for trade credit terms) between firms whose products were dereserved and a group of control firms whose products were never reserved/dereserved. In addition, I also find that for average firm, the price of firm's dereserved primary product also goes down by 8.6%. The incumbent supplier firms choose to extend trade credit at longer terms and also cut prices in anticipation of increased competition. This finding is robust to the inclusion of industry-year fixed effects, different sample periods, placebo samples and alternative definitions of treatment and control firms.

I then test how the choice of extending trade credit vis-à-vis price discounts as a pre-emptive strategic tool vary with the financial strength of supplier firm. Extensive literature predicts that "deep-pocketed firms" will attempt to drive financially constrained competitors out of business (Bolton and Scharfstein (1990), in particular, long payment terms impose substantial

liquidity risk on financially weaker suppliers, and force them into financial distress more often than they would if they were paid earlier (Barrot (2016)). Consistent with this view, I find that large and old firms (using size-age index proposed by Hadlock and Pierce (2010)) offer longer terms of credit, while financially constrained firms to opt for price discounts. To further confirm the results, I use a quasi-natural experiment i.e. establishment of Debt Recovery Tribunals (DRTs) in India (Gopalan, Mukherjee, and Singh (2016))—one that reduced the cost of legal enforcement of debt contracts in different states without any accompanying change in the underlying law— as a proxy for access to finance. I find similar results. I find that firms located in early DRT states increase trade credit by 18.9% (or an increase in term of trade credit by 17 days for the previous level of 73 days) and cut prices by 9.2% one year after the dereservation of its primary product compared to one year before. While firms in late DRT state do not change trade credit terms and cut prices by almost 40%. Also, I find that firms located in early DRT states finance the extra term on trade credit by altering maturity of bank debt.⁸ This is consistent with Parise (2015) who finds firms adjust debt maturity when they face entry threat from new competitors.

It is also important to understand why customer firms take trade credit rather than price discounts. Although I do not observe the customers of sample firms, I can look at the sample firms as a customer firms and test if they receive more credit when their suppliers face competition. In other words, I look at trade credit received by sample firms after dereservation of products used as their main input i.e. raw material. I find that when sample firms primary input dereserved, they receive more trade credit and price discounts. Interestingly, I find only financially constrained firms, identified using size-age index and establishment of DRTs in the state, take more trade credit, while unconstrained firms take price discounts.

Overall, my analysis contributes to a few strands of literature. First, my analysis is closely

⁸ This is consistent with the lower cost of long-term debt in early DRT states (Visaria (2009)). In my data I find that one year before the dereservation, average interest cost for long-term debt for median firm located in early DRT states is 8.8% p.a., while in late DRT states the cost is about 15% p.a.

related to Barrot (2016), who uses exogenous variation in trade credit supply to identify the causal effect of trade credit supply on industry dynamics. In contrast, I examine the other side of this interaction – I estimate the causal effect of entry threat on firms’ choice of extending trade credit. In this sense, our papers complement each other. Also, I provide comparisons between the choice of trade credit vis-à-vis price discounts when supplier/customer firm face financial constraints, thereby enrich our understanding how and when firms use trade credit as a strategic tool.

Second, my study also relates to a growing literature that documents how incumbent firms respond to the threat of entry. The theoretical literature in this domain discusses various strategic tools and rationales for *pre-emptive* actions (Dixit (1980)). On the empirical side, Frésard and Valta (2015) finds that investments by domestic US firms go down in response to the lower cost of entry for foreign firms. Parise (2015) and Goolsbee and Syverson (2008), document how incumbent carriers adjust debt maturity and prices, respectively, when the probability of future entry by low-cost air carriers in their route network dramatically increases. Cookson (2016) finds physical capacity expansion by low leverage incumbent US casino firms when they face entry threat. In this context, my study examines the role played by financial constraints for strategic adjustments in trade credit versus price.

Third, it relates to studies that examine the role of product market competition in corporate finance, especially to the few empirical papers relating trade credit to product market structures (e.g. Fisman and Raturi (2004), Hyndman and Serio (2010), or Barrot (2016)).

However, my study incrementally and substantially contributes to the literature above along four distinct dimensions. First, I rely on plausibly exogenous variations from the removal of entry barriers at the product level to cleanly identify the causal effect of product-market entry threats on firms’ supply of trade credit and product prices. Second, I concentrate on firms’ trade credit response to entry threats, as opposed to credit supply behavior following entry. This distinction is important because it allows me to examine the value of trade credit as a

pre-emptive strategic tool, without influence from other confounding factors associated with actual changes in competition (such as changes in firm financial position). Third, because I can observe the firm’s sales revenues across various products sold and quantity produced I can estimate the firm’s exact *exposure* to the entry threat. The availability of input-output data and prices further enhances the richness. In addition, the product-level sales data allows me to control for time-varying unobservable characteristics at a greater granularity, thereby aiding identification.⁹ Finally, the institutional setting allows me to examine *plausibly* exogenous shocks to the threat of entry across a broad spectrum of industries, rather than focusing on a single industry.

The remainder of the paper is organized as follows. Section 1 reviews the theory and evidence on trade credit and details the main hypotheses. Section 2 presents the dereservation reforms, which serve as the main source of identification in this paper. Section 3 presents the data and methodology. Section 4 describes the results. Section 5 provides robustness of results. Section 6 concludes.

1 Trade Credit Theories and Hypotheses

In this section I survey various trade credit theories and how they are related to product market competition. Then I use these theories to develop my hypotheses.

1.1 Trade Credit Theories

The trade credit literature has expanded in two directions: a financial motive approach and a product market competition approach.

The *financial motive* approach of trade credit explains how credit rationing affects the de-

⁹ I define firm’s product at five-digit National Industrial Classification (NIC) code. Therefore, I can control for time-varying unobservable characteristics at the four-digit code. There are 2,710 products, defined as five-digit National Industrial Classification (NIC) codes linked to 137 four-digit NIC industries across the 24 manufacturing sectors (two-digit NIC codes). In comparison, U.S. data used by Bernard, Redding, and Schott (2011) contain approximately 1,500 products, defined as five-digit Standard Industrial Classification (SIC) codes, across 455 four-digit SIC industries.

mand for trade credit.¹⁰ This approach has explored the motives of the “lender” and “borrower” of trade credit compared to other financial sources. It has focused on the problem of why a lender gives trade credit to a buyer, and why a buyer chose to “borrow” trade credit instead of some other type of borrowing. This approach implicitly assumes that financial motivation leads to trade credit provision.

A comprehensive survey of theories and empirical tests on *financial motive* for trade credit is done by Petersen and Rajan (1997). They use the SME data in the United States, and find that suppliers are inclined to lend to financially constrained customers. One interpretation of this result is that as suppliers have more information about their buyers, they offer goods on credit. Burkart and Ellingsen (2004) suggest that suppliers have an informational advantage which banks do not have. They hypothesize that it is typically less profitable for an opportunistic borrower to divert inputs than to divert cash, which increases the advantage of suppliers over banks in lending to their clients. The advantage suppliers have over external financiers may also be nested in the nature of the underlying product. Fabbri and Menichini (2010) theorize that liquidation advantages of the supplier to other financing sources allow them to provide credit. This comparative advantage is more pronounced for differentiated goods because these are often tailored to the needs of few customers.

The *product market competition* approach concentrates on how competition in the product market affects trade credit provision. The literature presents contrasting evidence in this context. On the one hand, McMillan and Woodruff (1999) find that the presence and number of competitors within a 1-km area lower trade credit provision to customers. Similarly, Johnson, McMillan, and Woodruff (2002) also find that trade credit provision is lower when there exist more than 5 rivals within 1 km. Their results suggest that competition prevents suppliers from giving credit for avoiding risk. On the other hand, subsequent studies find an opposite result.

¹⁰ Some other theories of trade credit suggest that trade credit reduces transaction costs (Ferris (1981)), allows price discrimination between customers with different credit worthiness (Brennan, Maksimovics, and Zechner (1988)), provides a warranty for quality when customers cannot observe product characteristics (Long, Malitz, and Ravid (1993)), and fosters long-term relations with the customer (Wilson and Summers (2002)).

Fisman and Raturi (2004) document that the monopoly power of the supplier is negatively associated with credit provision, which counters the assertions of previous studies that monopoly power facilitates the provision of credit because monopolists are better able to enforce payment. Later, Hyndman and Serio (2010) show that the relationship between trade credit provision and supplier's market power is not linear but follows an inverted-U shape. A monopolist supplier often prefers to sell only in cash, which is zero trade credit. Once competition starts, trade credit grows with the number of competitors. Hyndman and Serio (2010) argue that this happens as Bertrand price competition in the cash market pushes up the price of cash, and thus new entrants can only offer trade credit given the product market competition. With the intensification of competition, problems of commitment on trade credit repayment and decisions on credit provision become less important. However, enforcement becomes constrained as the number of competitors increase and outgrows a certain limit.

In this paper I combine both the motives of trade credit and show how firm's choice of trade credit vis-à-vis price as a strategic tool vary with financial constraints. The above *product market competition* theories all argue how industry structure affects trade credit. Barrot (2016) documents the reverse: how trade credit provision affects industry dynamics. He finds that restriction on trade credit provision affects the entry/exit rates of trucking firms in France. He concludes that longer terms extended by financially stronger firms might thus act as a barrier to entry and prevent their constrained, -yet potentially efficient- rivals, from entering, expanding, and surviving in the industry. However, no one has explored whether incumbents firms use trade credit or price or both, *ex-ante*, when they face entry threat from competitors.

1.2 Hypotheses Development

The literature above suggests that incumbent supplier firms should extend more trade credit before the potential entrant starts manufacturing the product. In other words, incumbent supplier firms should lengthen the payment terms of trade credit to their customers when

they face entry threat due to the removal of entry barriers. Trade credit raises barriers for the potential entrant. However, if incumbents can easily adjust trade credit terms after the entry, then longer payment terms to customers may not serve as a credible strategic tool for entry deterrence. Barrot (2016) finds long payment terms (measured as the ratio of accounts receivable/sales) impose a substantial *liquidity risk* on financially weaker suppliers, and force them into financial distress more often than they would if they were paid earlier. This suggests that increased trade credit *may not be easy to adjust back*. Therefore, extending trade credit is an effective signal that is costly and credible. This suggests that effective increase in competition encourages incumbent firms to increase trade credit. While at the same time the limit-pricing (Salop (1979) and Milgrom and Roberts (1982)) theories suggests that similar benefits can be given to customers by offering a lower price and that can also help deter the entry. This suggests that incumbent firms may use price discounts as a strategic tool.

H1 Price vs. Trade Credit Hypothesis: With increased threat of entry, incumbent firms *increase trade credit* to their customers to prevent them from switching to the new suppliers. Or, they *cut down prices* or both.

A natural extension relates to the incumbent supplier firms' ability to finance trade credit. A large literature shows that "deep-pocketed firms" will attempt to drive financially constrained competitors out of business (see e.g. Telser (1966), Bolton and Scharfstein (1990)). Therefore, when incumbent supplier firms expect greater competition in future, *ex-ante*, financially strong incumbent firms may extend more trade credit. While financially constrained firms may use limit-pricing as a strategic tool.

H2: Supplier's Financial Strength Hypothesis: With increased threat of entry, financially strong (weak) incumbent supplier firms *increase trade credit (decrease prices)* to influence the entry decision of the potential entrants and prevent customers from switching to new suppliers.

At the same time when a supplier offers more credit, customers firms will take it only if they

are financially constraint otherwise they take price discount. Financially constrained customer firms, who value a dollar of credit more than a dollar, take more credit from their suppliers, when supplier firms face greater product market competition.

H3: Customer's Financial Strength Hypothesis: With increased threat of entry for supplier firms, financially weak (strong) customers *take more trade credit (ask for price discounts)* from supplier firms.

2 Dereservation Reform

The Indian Government has a long history of promoting small-scale industries. Starting in 1960, the Government of India reserved a large number of manufactured goods for exclusive production by small scale firms. Hundreds of products across the manufacturing sector were only allowed to be produced by small scale firms, insulating them from competition.¹¹

It was argued that small establishments producing labor-intensive goods would absorb the abundant labor supply present in an underdeveloped country. However, in official documents there is no clear criterion for the selection of goods to be reserved. In addition, there is no mention of other criterion for reserving specific goods say, based on optimal capital to labor ratios (which are difficult to ascertain in the first place). For example, in clothing, cotton and woolen socks, scarves, clothes and vests were reserved, while linen, jute or hemp products were not reserved. This suggests a high degree of substitutability between reserved and non-reserved clothing items. The types of products on the reserved list were varied, spanning many industrial sectors such as food, chemicals, electronics and textiles. Within the small-scale sector, the output share of reserved products was approximately 30% in 1987. Overall, reserved products constituted about 12% of Indian manufacturing output. There was considerable heterogeneity

¹¹ The Indian government defines a small-scale firm according to the cumulative amount of investment in plant and machinery. This means that all the plants with a level of capital below a limit set by the government are considered “small” and, therefore they are allowed to produce reserved goods. Such limit has been changing over time. It started at (0.5 million INR in 1960 and has been periodically adjusted upward using inflation (García-Santana and Pijoan-Mas (2014)).

across industry sectors with reserved products forming 80% of output in hosiery and garments, 57% in certain wood products, and a negligible fraction in textiles.¹²

Despite India's liberalization of a variety of industrial and trade policies in 1991, the reservation of products for small scale sector remained in force until the late 1990s. As of July 1991, there were around 800 items which constitute around 1,024 products in the reserved category.¹³ As per this policy, only small scale firms could manufacture these products. Large/Medium sector firms that already made the reserved products were permitted to continue to manufacture within the reserved product category, but their output was capped at current levels. They could expand the capacity only if they agree to export a minimum of 75% of their production from fresh capacity. This regulation induced heterogeneity in the reserved sector which includes small firms and export-oriented large firms.

The Advisory Committee on Reservation recognized growing concerns about small scale sector policies that followed the 1991 trade liberalization. The small scale firms had to compete with imported goods, and large undertakings might be able to exercise monopoly power in the market for reserved goods as most other producers would be small. Moreover, growing consumer demand for high-quality goods, and ongoing technological progress, made it more difficult to produce many items in this sector. The Advisory Committee therefore appointed a special committee to reconsider the list of reserved items in 1995 (MSME (2007)). Based on recommendations from this committee, most of the 1,024 products were dereserved starting in 1997 (Figure 1).¹⁴

The identification of the threat of entry in this paper comes from rapid and complete dismantling of the reservation policy. Large scale dereservation started slowly in 1997 (15 products) and picked up in 2002 (51 products). From 2003 to 2008, approximately 100 to

¹² An expert committee, Abid Hussain Committee, constituted in 1997 to review small scale industry in India's post liberalization period states, "the choice of products for reservation was necessarily arbitrary" (Hussain (1997)).

¹³ As per Notification S.O 477(E) issued by Ministry of Micro, small and medium enterprises (MSME), Govt. of India. <http://www.dcmsme.gov.in/publications/circulars/477E.pdf>

¹⁴ For a more detailed description on the process of dereservation: <http://dcmsme.gov.in/publications/reserveditems/itemrese.htm>.

250 products were dereserved each year. Finally, in April 2015, the last 20 products were removed from the reservation list. Appendix A (table A1) shows sample notification about list of items dereserved in 1997.¹⁵ For example, 15 products were dereserved in 1997 which included ice-cream, poultry feed, hair dryers, ash tray etc.

As evident from Appendix A (Table A2), the dereservation of products are staggered across time and industries. To give an overview for variation across industries, from 2003 to 2008 a total of 809 products over 24 (2-digit NIC) manufacturing industries were dereserved of which chemical products contributed the most (30%). For example, products in Basic Metals (2 digit NIC Code =24), were never dereserved during this period. However, 107(13.23% of total 809 products) products in Fabricated Metals (2 digit NIC Code =25), were dereserved. Similarly, 13(1.61% of total) wooden products (2 digit NIC Code =16), were dereserved while 11(1.36% of total) furniture products (2 digit NIC Code =31), were dereserved.

Similar staggered variation in dereservation is also observed across industries over time. As discussed before, products in Basic Metals industry were never reserved/dereserved while products in Fabricated Metals industry were dereserved in a staggered manner i.e. 0%(2003), 22%(2004), 0%(2005), 42%(2006), 31%(2007) and 5%(2008). Similar staggered variation is observed in other industries over time and Appendix A (Table A3) gives more information about timing of dereservation across industries over time.

Was dereservation really random? One could be concerned about the potential anticipation of the dereservation policy. Anecdotal evidence and discussions suggests that reservation of products and the timing of their dereservation appear to be arbitrary in nature and varied across different industries. As in the case of reservation, the process of dereservation is also not well-understood. A reading of government documents, reports and media does not give clear-cut reasons as to why certain products were initially reserved or dereserved at certain times.

Martin, Nataraj, and Harrison (2016) and Tewari and Wilde (2014) document that a conclu-

¹⁵ All the notifications about dereservation are available at <http://www.dcmsme.gov.in/publications/notiissforresderes.html>

sive and definitive account of the dereservation process is not available. Explanations range from competition from imports, technology requirements, need to comply with regulations, no benefit to small producers or availability of unreserved substitute products. A product’s path to dereservation tends to be lengthy and circuitous. A product is identified as a dereservation candidate by a ministry or industry players (including manufacturers of reserved products themselves who find the investment ceiling constraining). Once identified, a series of meetings between “stakeholders” (such as trade associations or small firm groups and officials) takes place. After review from a chain of bureaucrats, the dereservation of a product is signed into law by the central government minister. Qualitative support for the above arguments on the nature of reservation and dereservation is reflected in the extent of reservation/dereservation both across and within product categories.

As an example, among various oils, many oils that were never reserved, and several (like sesame oil, mustard oil and rapeseed oil) were reserved. However, they were dereserved at different point in times such as hair oil was dereserved in 2003, sesame oil in 2008, and mustard oil in 2015. Similarly, among leather products, leather shoes were dereserved in 2001 while leather slippers were dereserved in 2003. It is difficult to think of some systematic and endogenous rationalization in this cross-section and time-wise pattern of reservations.

3 Data and Methodology

3.1 Firm Characteristics

I compile a firm-level panel data set that spans from 1990 to 2013 from Prowess_{dx} database, collected by the Centre for Monitoring the Indian Economy (CMIE).¹⁶ The database contains information primarily from the income statements and balance sheets of about 34,197 publicly

¹⁶ Prowess_{dx} (<http://prowessdx.cmie.com>), is special data extraction interface for academicians, similar to Prowess. The data set has been used by number of prior studies on Indian firms, including Bertrand, Mehta, and Mullainathan (2002), Gopalan, Nanda, and Seru (2007), Lilienfeld-Toal, Mookherjee, and Visaria (2012) and Gopalan, Mukherjee, and Singh (2016).

listed and private companies, of which 10,724 are in the manufacturing sector. This database is a firm-level panel that also records detailed annual information on firms' product mix.¹⁷ Furthermore, for each product manufactured by the firm, the data set provides the value of sales, quantity, and units. With this data one can easily create price index for each product for a given firm-year. In addition to output, the data set provide information on all the materials used to manufacture the reported products. The Prowess_{dx} is therefore particularly well suited for understanding how firms adjust their financial policies and price policies when their product lines are under threat of competition. Also, with the availability of the input data, one can look at the firms behavior as a customer. The definition of a product is based on the CMIE's internal product classification. I complement the data on firm product mix with measures on product reservation policy at the 5-digit NIC (similar to SIC of US) product level. I follow Goldberg, Khandelwal, Pavcnik, and Topalova (2010) to clean up the product-level database. Appendix A provides more details on product data. I follow the sample firms from beginning of the year 1990 to end of 2013. All financial variables are inflation adjusted using Wholesale Price Index (WPI) at 2004-2005 prices. I also correct for change in financial reporting year. To make sure that my estimates are not influenced by industry-level trade liberalization policy, which was completed by 1997, in robustness checks I start sample from the beginning of year 1997. Thus my estimates are robust to sample selection period.¹⁸ Since the reservation policy was meant for only manufacturing firms in India, I keep only non-services, and non-government firms in my sample.¹⁹

¹⁷ Indian firms are required by the 1956 Companies Act to disclose product-level information on capacities, production, and sales in their annual reports. The CMIE compiles these detailed quantitative data.

¹⁸ Major changes in policies vis-à-vis foreign investment occurred in the early 1990s, and then stalled during the period of dereservation reform. Nataraj (2011) shows that tariffs were largely harmonized across industries by the late 1990s, so even though there were some reductions during the 2000s the variation in tariff rates across product types had fallen dramatically by the start of the 1997.

¹⁹Unlike Martin, Nataraj, and Harrison (2016) who use ASI data, I rely on Prowess database due to the joint availability of information on trade credit, price data, input and output data for a given firm. In addition, ASI provides establishment level information while in this study I am more interested in firm-level strategic actions. I understand that I may not be capturing very small and micro enterprises who were manufacturing the reserved products, but the data captures relatively medium and large sized incumbent firms who were capacity constrained. And with removal of entry barrier, it remains interesting to see how capacity constrained incumbents respond. Prowess data do not provide much information on employment and wages, which was the

3.2 Measure of Trade Credit Supply

Following Petersen and Rajan (1997), I proxy payment terms by the amount of accounts receivables on firms' balance sheets as recorded at the end of their year of operations *scaled by* total cumulative sales in the year (Accounts Receivables/Sales). A higher ratio implies significant amount of cash is tied up. In other words, an increase in accounts receivable to sales ratio from one year to the next indicates that investment in accounts receivable is growing more rapidly than sales.

This widely used measure of trade credit provision is a rough proxy for actual payment terms. A seasonal business experiences a large part of its annual sales in a particular part of the year and hence this ratio can be seasonal. However, it is unlikely that this seasonality in the sample changes dramatically following the dereservation reform. In addition, inclusion of industry-year fixed effects at more granular level helps absorb seasonality within industries across time. Similarly, this ratio may mechanically overestimate average payment terms in periods of growth and underestimate during downturns. This makes it important to have a right control group. The staggered nature of reform helps me use the pre-reform period observations of treatment firms as an appropriate control group and thus aids identification of the effect of entry threat on payment terms.

In addition to Accounts Receivables/Sales, I estimate my results using $\text{Log}(1 + \text{Accounts Receivables})$ to further confirm the findings. To make sure that results are not effected by outliers, I winsorise all the ratios at 1% and drop observations with ratio of accounts receivables to sales greater than 1.

3.3 Summary Statistics

As reported in Table 1, median Accounts Receivables/Sales ratio is about 0.16 with mean of 0.19, equivalent to average payment period of about 70 days ($=0.19 \times 365$ days). The median

interest of Martin, Nataraj, and Harrison (2016) study.

firm's price of output measured as *Log (Primary Product Price Index)*, is 0.46, which translates to price index of 1.25. For average firm, 18% of assets are financed using trade credit. The median firm size, measured as *Total Assets* in my sample is \$11.17 Mn., which translates into a book value of total assets amounting to INR 503 million at 2004-2005 prices (with \$1=Rs.45, exchange rate). The 10th percentile and 25th percentile firms have book value of total assets amounting to INR 77 million and INR 180 million, while the minimum is about INR 0.88 million. Firms in India have little cash on their balance sheet as seen by the mean value of cash holdings of 6%. Firms in my sample are profitable, as seen from the mean value of profitability (EBIT/Assets) of 12%. On average, sales growth is about 14%.

In my sample, firm produces a median of two products, with mean of 2.7 products. The inter-quartile range for sales revenue from the primary product varies from 70% to 99%.

3.4 Methodology

To identify the incumbent supplier firms that are affected by threat of entry, I define treatment as the elimination of small-scale reservation on the incumbent firm's primary product in the NIC-5 digit category. Firm's primary product is the product that has the maximum sales revenue in the immediately preceding year of dereservation. I start with a difference-in-differences (DID) equation of the following form for firm i in year t :

$$y_{it} = \alpha_0 + \sum_{k=-2}^{-10} \beta_k \text{Pre-Dereservation}(k)_{it} + \sum_{k=0}^{10} \beta_k \text{Dereservation}(k)_{it} + \alpha_i + \alpha_t + \varepsilon_{it} \quad (1)$$

The dependent variable y_{it} is defined as the *accounts receivables scaled by sales* of firm i at time t . $\text{Pre-Dereservation}(k)_{it}$ ($\text{Dereservation}(k)_{it}$) is a dummy variable that takes a value one if it is k years before (after) the incumbent firm's primary reserved product has been dereserved. $\text{Dereservation}(k)_{it}$ dummy identifies the group of firms under threat of entry. In other specifications I replace dummy with the exposure of incumbent firm to entry threat by using the proportion of total sales revenue from de-reserved product in pre-dereservation period.

An incumbent firm is defined as a firm that has ever made a reserved product before it was dereserved.²⁰ In certain specifications I include all firms – even those that do not help to identify β_k because they are not affected by the reservation policy – because these firms help to identify the secular year trends in firms’ financial policies. Also, *Pre-Dereservation (-10)* equals one if it is ten or more years before dereservation of product and *Dereservation (+10)* equals one if it is ten or more years after dereservation.²¹ The model is fully saturated with the year immediately before the dereservation as the excluded category. Therefore, the coefficients on *Pre-Dereservation(-k)* (*Dereservation(k)*) compare the level of the dependent variable ‘k’ years before (after) the dereservation to the year immediately before its dereservation. The inclusion of firm fixed effects, α_i , ensures that each indicator is estimated using only within firm variation in the dependent variable, and time dummies, α_t controls for time trends. I recognize that incumbent supplier firms previously manufacturing reserved products may have secular time trends that differ from non-incumbents. To control for this, in various specifications I include interaction between the year and incumbent dummies. The identification at firm-level helps me to absorb various time-varying industry-level heterogeneity. Standard errors are corrected for heteroscedasticity and auto-correlation, and clustered at NIC5-digit product level.

The removal of entry barriers should satisfy one requirement to be “exogenous” i.e. the occurrence of these events should be unrelated to individual firms’ strategic decision of extending trade credit. As discussed before, reservation of products and their dereservation appear to be random in nature and varied across different industries.

In addition, I employ Cox survival model to investigate whether, if any, observable differences in average firm characteristics across industries predict the timing of dereservation. The sample for this regression spans 1990-2009 and includes the industries that were never reserved. In ad-

²⁰ Results are robust to alternative definitions of incumbent firms. Table 9, column (5) reports results where incumbent is defined as a firm that made a reserved product three years before it was dereserved as its primary product.

²¹ Since coefficient on *Pre-Dereservation (-10)/Dereservation (+10)* represent effect for more than 1 year, I do not report these coefficients.

dition, I include average characteristics of firms in each of the industries prior to dereservation. Specifically, I estimate industry-wide averages (at NIC 5 digit level) of these firm characteristics over the period 1990-1996 i.e. six years before the first dereservation announcement. In alternative specifications, I include the average value of Accounts Receivables/Sales, Firm Size, Cash Holdings, Profitability and Sales Growth. Table 2 reports the results from this analysis. I find that the coefficient on all the firm characteristics are insignificant in all the specifications, after including industry-fixed effects to control for industry-specific unobservables.

These results confirm that there is no systematic difference in the average characteristics between the firms in early and late dereserved industries. In further analysis below, I conduct various robustness checks and placebo tests to verify the validity of results. Although, it is difficult to think of some systematic and endogenous rationalization in this cross-section and time-wise pattern of reservation, I cannot entirely rule out this possibility.

Next, for dereservation to be a plausible variation for threat of entry, it should be related to market competition. Also, dereservation should not affect the industry concentration immediately. I use different measures of industry competition i.e. entry rate, product prices and market concentration index(HHI). I estimate *entry rate* as ratio of number of entrants to number of incumbents in five-digit industry for a given year. While for *industry level product price index*, I create firm-level price index for each product and take weighted average over all the products in a given year for each five-digit industry. For firm-level price index I use first observed quantity sold as a numeraire and track the price change. Finally, I measure market competition/concentration by Herfindahl–Hirschman Index (HHI). I estimate regression model similar to equation (1) with (a) Entry Rate, (b) Log(Average Product Price Index) and (c)Log(HHI) (measured at 5-digit NIC product-level) as dependent variable. Here, the control group includes pre-event observations of reserved products that got dereserved later and never reserved/dereserved products. Thus, a difference-in-difference estimate suggests how market competition is affected in reserved product category compared to the control group.

Figure 2 plots the regression estimates and provides suggestive evidence on non-existence of pre-trends in entry rate/product prices/market concentration. Though not a perfect test, this shows that changes in competition did not really affect the dereservation decisions and therefore seem exogenous. The entry rate increases significantly after dereservation and average industry price index goes down by almost 18% four years after the dereservation. In addition, I find that market concentration goes down significantly by 10%, about 5-6 years after the dereservation. This suggests that competition within dereserved industries increased significantly subsequent to the removal of entry barriers. This is consistent with the fact that, new entrants take time to setup plant and/or expand capacity to manufacture dereserved products. These new entrants may start manufacturing within few years of dereservation, but will start affecting market concentration only after 5-6 years as they begin to capture significant market share. This non-immediate effect of dereservation on market concentration helps me identify *threat of entry* from *actual entry*. This suggests that dereservation had real effects on product market competition and threat of entry thereby affecting firm's policies. In effect, dereservation seems like a good instrument for threat of entry.

4 Results

4.1 Main Results

I start my analysis in this section by plotting regression coefficients from my baseline specification, equation (1) for dereserved products only.²² Figure 3 shows the difference-in-differences estimate of trade credit granted (*accounts receivables over sales*) in event time, and the 1% confidence interval around this difference. Although, I present the evidence till 9 years after the dereservation event but I'm interested in short window of 5 years i.e. only *threat of entry* period instead of *actual entry*. As observed in Figure 2 (C) and discussions above, I find a

²² Note that here the control group includes pre-event observations of firms producing reserved products that were dereserved later. It does not include never treated firms; so I do not include incumbent-year fixed effects.

significant change in market concentration only after 5-6 years. This short window of 5 years around the event, therefore, captures the average causal effect of the pre-emptive action.

The patterns in the figure are striking. First, there are no discernible pre-trends in my data – the difference between the treatment and control groups is statistically insignificant in the five years prior to the dereservation. Second, with dereservation, there is a clear jump in trade credit granted (measured as accounts receivables over sales) one year after the dereservation (statistically significant). Also, the effect stabilizes after three years of dereservation. In further analysis, for brevity, in tables I report results for coefficients on *Dereservation* ($k=-3$) and *Dereservation* ($k=-2$) and *Dereservation* ($k=0$) to *Dereservation* ($k=+6$).

Table 3, column (1) presents the above results in tabular form. In column (2), I include firms whose products were never reserved/dereserved and include an interaction between the year and incumbent dummies (incumbent $\times$ year fixed effects). Similar to column (1), in this specification as well, I find that the coefficients on *Dereservation* ($k=-3$) and *Dereservation* ($k=-2$) are statistically insignificant. This further confirms dereservation process is exogenous to firm’s strategic decision to change trade credit. I find that the coefficients on *Dereservation* ($k=0$) through *Dereservation* ($k=+6$) are positive and significant in column (1) and column (2). This shows that after the dereservation of products, investment in accounts receivable is growing more rapidly than sales. The positive and significant coefficient of *Dereservation* ($k=+1$) implies that from one year before to one year after the removal of entry barrier, ratio of accounts receivable to sales of treated firms increase by 8.9 % (0.017 divided by pre-dereservation mean of 0.19) relative to the ratio of control firms.²³

4.1.1 Placebo Dereservation

In order to alleviate concerns about my results being driven by industry conditions, I employ a placebo test. This test would expect my results to remain even if I replaced my

²³The observations in regressions do not match the summary statistics because *reghdfe* command automatically drop singletons to correctly estimate the clustered standard errors (Correia (2015)). All the results are qualitatively similar if one keeps the singletons, and are available on request.

Dereservation dummy as one for firms manufacturing similar products which were never reserved/dereserved. Since these products were likely to face similar demand/supply shocks (but did not get reserved/deserved), any effects of similar magnitude as earlier would raise concerns about dereservation as a valid instrument for entry threat.

In this sub-section I provide results from a randomized assignment of dereservation years to similar products, keeping the distribution of the event years unchanged. For each product dereserved (defined at 5-digit level) in the sample, I create a pseudo dereservation sample, where instead of the product that actually was dereserved I assign it to one randomly selected similar product (i.e. at the same 4-digit level). In each of the pseudo dereservation samples I run regression as in Table 3, column (2) and save the relevant coefficients. I repeat this procedure 5,000 times to obtain a (non-parametric) distribution of coefficients obtained from the placebo dereservation samples.

In Figure 4, I plot these distributions of the coefficients. The black line embedded in the graphs represents the regression coefficient obtained using the actual data. For example, figure shows that the coefficient for *Dereservation* ($k=+1$) in my placebo sample lie in the tails of the placebo distribution (p-value<0.001). Thus, unobserved industry shocks affecting trade credit supply are not driving my results and bolsters my identifying assumption.

4.1.2 Exposure to Dereservation

In this sub-section, I measure the exposure of firms to dereservation as proxied by sales contribution of affected product to firm's total sales. In general, firms produce more than one product to stay competitive in a market. In fact, Goldberg, Khandelwal, Pavcnik, and Topalova (2010) finds that Indian multiproduct firms are quite similar to their counterparts in US manufacturing firms studied by Bernard, Redding, and Schott (2011). Like, Goldberg, Khandelwal, Pavcnik, and Topalova (2010), I find that the typical firm in my sample produces two products and the average sales revenue from the primary product varies from 70% to 95%.

I measure exposure of incumbent firm to threat of entry with proportion of its sales revenue

from dereserved product in pre-dereservation period and report the results in Table 3, column (3). I find that incumbent firms with greater exposure to threat of entry extend more trade credit. In addition, accounts receivables over sales increases by 2 percentage points (about 8.2% increase compare to pre-dereservation mean of 0.19, with 0.81 exposure i.e. $0.02 \times 0.81 / 0.19$) four years after the dereservation of the firm’s primary product. Column (4) is like column (3) except with $\text{Log}(1 + \text{Accounts Receivables})$ as dependent variable, and I find similar results

4.1.3 Industry Unobservables

Next, one might be concerned about trade credit terms varying across industries over time and dereservation may be correlated with time-varying industry unobservables i.e. demand/supply conditions. To this effect, I control for unobserved heterogeneity at industry-level. In table 3, column (5), I carefully interact the 4-digit industry-year fixed effects for both treatment (incumbent) and never-treated control group. The effect of dereservation remains positive and statistically significant. After including industry-year fixed effects at 4-digit level, which is possible because dereservations are measured at the 5-digit level, I find that from one year before to one year after the removal of entry barrier, the ratio of accounts receivable to sales of treated firms increase by 2.4 percentage points relative to the ratio of control firms. This effect is economically large; the jump in trade credit represents a relative increase of about 10.2% from the pre-event level.

In column (6), table 3 report results after including control variables from prior literature (Petersen and Rajan (1997) and Barrot (2016)). To make sure inclusion of the control variables does not bias estimates, I lag the control variables by a year. I include one year lagged value of firm size, cash holdings, profitability and sales growth as controls. I find that inclusion of the control variables has a negligible effect on the size of the coefficients of interest. For example, the coefficient on *Dereservation* ($k=+1$) changes from 0.024 in column (5) of Table 3 to 0.026 in column (6) of Table 3. This effect is economically large; the jump in trade credit represents a relative increase of about 11% from the pre-event level.

These results are consistent with hypothesis *H1* i.e. the strategic use of extending trade credit is to defend market share. With expected competition, incumbent firms extend more trade credit before the potential entrant starts manufacturing the product (Fisman and Raturi (2004), Hyndman and Serio (2010) and Barrot (2016)). This pre-emptive action to increase accounts receivables may serve as entry barrier for potential entrant.²⁴. Hence, the potential entrant requires not only technological and organizational expertise, but also the capacity to extend credit to the customer base.

4.1.4 Price of Output

As discussed above that incumbent firms, in order to defend their market share, extend more credit to their customers. But, do they simultaneously reduce prices or keep them unchanged? In my sample, I can observe the quantities sold and sales revenue generated from each product for a given firm-year. Using this data, I create firm-level product price index. Firstly, I use first observed quantity sold as a numeraire and track the price change. I identify sample firm's primary product each year based on total sales revenue from the product portfolio.

Table 4 reports the results for difference-in-differences regressions for sample firm's Primary Product Price Index. Column (1) report results with dereservation dummies and incumbent $\times$ 4-digit product $\times$ year fixed effects. The results are striking. The negative and significant coefficient of *Dereservation* ($k=+1$) implies that from one year before to one year after the removal of entry barrier, price of treated firms' primary product decrease by 9.4 % relative to the control firms. While for next three years price differences were not significant for treated and control group. Interestingly, five year after the removal of entry barrier i.e when we start to see entrants gaining market share (Figure 2 (C), log (HHI) significantly lower after 5–6 years of dereservation), I find cut in prices by almost 17.5%. It appears that incumbent firm's adjust prices in step-function. In column (2) I replace dereservation dummies with the proportion of total sales revenue from de-reserved product in pre-dereservation period. While in column (3)

²⁴I discuss the effectiveness of trade credit as a strategic tool in section 4.5

I add firm controls, as in table 3. Finally, in column (4)-(6), I adjust $\log(\text{Primary Product Price Index})$ for inflation using wholesale price index. The results are consistent i.e. I find firms do cut prices as a pre-emptive action. For average firm manufacturing dereserved as its main product one year before the dereservation, lowers prices by 8.6% (0.81×0.107) one year after the dereservation.

4.2 Access to Finance

It's important to understand whether firm's choice of strategic action may be influenced by access to finance/financial strength. I test hypothesis *H2* i.e. how incumbent firm's response to threat of entry varies with their financial strength/access to finance. I follow Hadlock and Pierce (2010) and use firm size-age (SA) index a proxy for financial strength. In addition I use a quasi-natural experiment i.e. establishment of Debt Recovery Tribunals (DRTs) in India (Gopalan, Mukherjee, and Singh (2016)), as proxy for access to finance and report results in table 5.

I re-estimate column (6) of table 3, my baseline specification, with interaction terms *Dereservation* (k) $\times$ *Low SA* and *Dereservation* (k) $\times$ *High SA*, where *Low SA* is a dummy variable that identifies firms with below-median Size-Age Index in the year prior to dereservation of its primary product and report results in Panel A of Table 5. I do this for both *accounts receivables over sales* and $\log(\text{Primary Product Price Index})$ as dependent variables. The evidence suggests that firms with less financial constraints increase *accounts receivables over sales* and while firms with lesser financial strength cut prices.

Further to confirm the results, I use establishment of Debt Recovery Tribunals (DRTs) in India—one that reduced the cost of legal enforcement of debt contracts in different states without any accompanying change in the underlying law—as proxy of access to finance. Various studies show that the lower enforcement costs are positively associated with the size of new loans and debt maturity, while negatively related to interest costs (Lilienfeld-Toal, Mookherjee,

and Visaria (2012), Gopalan, Mukherjee, and Singh (2016) and Visaria (2009)).

Like Panel A, Table 5, I re-estimate column (6) of table 3 with *Late DRT* and *Early DRT* as interaction terms, where *Early DRT* identify the state of firm’s registered office where DRTs were operational for at least five years in the year prior to dereservation of its primary product. Table 5, Panel B report the results. I find that firms located in early DRT states were quick in responding to entry threat by increasing credit offered to customers while incumbent firms located in late DRT states rely on lower price. I find that firms located in early DRT states increase trade credit by 18.9% (or increase in term of credit by 17 days for previous level of 73 days) and cut price by 9.2% one year after the dereservation of its main product compared to one year before. While firms in late DRT state do not change trade credit terms and cut prices by almost 40%.

Overall evidence supports the hypothesis that firm’s better access to financial market use trade credit as strategic tool while other firms rely on limit-pricing. These findings are consistent with the large literature that explains “deep-pocketed firms” attempting to drive financially constrained competitors out of business (see e.g. Telser (1966), Bolton and Scharfstein (1990)), while here we identify the exact tool i.e. offering more credit.

4.3 Financing Trade Credit

In last sub-section, I find that firms with better access to financial markets use trade credit as a strategic tool. Its important to understand how firm finance this excess credit offered to customers.

I estimate column (6) of table 3 for different dependent variables. Column (1) -(4) report results with (a) Leverage, (b) Cash Holdings (scaled by total assets), (c) Trade Payable (scaled by assets) and (d) Debt Maturity (ratio of long-term debt to total debt), respectively. I find no change in leverage, cash holdings or debt maturity while trade payable i.e. credit from supplier appears to increase. When I partition this based on firm’s access to financial market

using DRT, I find no difference between the two groups for leverage, cash holding or trade payable and therefore do not report here. Interestingly, I find that firm's are adjusting their debt maturity structure significantly in order to finance that longer terms offered to customers after dereservation.

4.4 Demand for Trade Credit

In this sub-section, I test hypothesis $H3$ i.e. how incumbent's response to threat of entry varies with financial strength of customers. Since I don't observe the customers of given firms, I try to test this hypothesis looking at the sample firm's as customers. I test whether incumbent firms whose main input is dereserved i.e input market become more competitive, receive more credit or lower price or both. Also, how access to finance affect these results.

4.4.1 Trade Credit Received

Table 7, Panel A report results for trade credit received as dependent variable. I measure trade credit received as *accounts payable/total assets*. Column (1) report results where control group includes only pre-event observations of firms using reserved products that were dereserved later (incumbents). While, in column (2) I include firms whose primary inputs were never reserved/dereserved and include an interaction between the year and incumbent dummies (incumbent $\times$ year fixed effects). Results from both columns suggest that incumbent firm's whose main input is dereserved receive more more credit from their suppliers. The positive and significant coefficient of *Dereservation* ($k=+1$) implies that from one year before to one year after the removal of entry barrier in firm's main input market, ratio of accounts payable to assets of treated firms increase by 8.8 % (0.016 divided by pre-dereservation mean of 0.18) relative to the ratio of control firms.

Column (3), to estimate exposure of firm to dereservation, I replace dereservation dummies with the proportion of total raw material expenditure from de-reserved product in pre-dereservation period. Column (4) is like column (3) except with $\text{Log}(1+\text{Accounts Payable})$ as

dependent variable. Column (5) is like column (3) except with Accounts Payable scaled by Cost of Goods Sold as dependent variable. Column (6) reports results where to column(3), I further add one year lagged value of firm size (measured as log sales), cash holdings, profitability and sales growth as controls. All regression are with incumbent $\times$ 4-digit product $\times$ year fixed effects (except column (1)).

Overall evidence suggest that incumbent firms receive more credit from their supplier after dereservation of its main input.

4.4.2 Price of Inputs

Similar to price of output, in table 4, I create primary input price index and follow the same specifications. For inputs, I find that price index for firm's primary inputs goes down by 11.3% to 12.9% in different specifications. The negative and significant coefficient of *Dereservation* ($k=+1$) in column (6) of table 7, Panel B, implies that from one year before to one year after the removal of entry barrier in firm's main input market, Primary Input Price Index of treated firms decrease crease by 11.3 % (exposure of 0.71 for main input, so effect is about 0.71×0.157) relative to the ratio of control firms.

4.4.3 Customer's Access to Finance

It's important to understand whether results vary by access to finance/financial strength of customer firm. I test hypothesis $H3$ i.e. how incumbent firm's response to threat of entry varies with customers financial strength/access to finance. Like table 5, here I partition the above results based on financial strength proxies i.e. Size-Age index and establishment of DRT in firm's registered office state. Panel C and D report results, respectively. The results suggest that firms with financial constraints i.e. Low SA and firms located in Late DRT states receive more credit while unconstrained firms take price discounts.

Overall findings in this sub-section suggests that financial constraints do impact the choice of strategic tool when firms face entry threat from new competitors. Firms with deeper pockets

offer more credit and firms with financial constraints take more credit, when bargaining power shifts from supplier to customer. While, unconstrained customer firms demand lower prices when their supplier face more competition.

4.5 Entry Deterrence

Finally, in this sub-section, I test the hypothesis whether firms that tried to defend their market share by increasing trade credit were successful or not. I look at the entry rates in five-digit industries separately for firms that chose to increase trade credit after dereservation and those that do not and report the results in Table 8.

I measure entry rate as ratio of number of entrants to number of incumbents in five-digit industry for a given year. Then I look at the change in entry rates after dereservation for group of firms which have increased (decreased) trade credit in the year and one year after the dereservation. I find that entry rates have increased after the dereservation for both groups but the increase is lower for the group that choose to increase trade credit in the year and one year after the dereservation. For, $\text{Dereservation}(k=+2)$, coefficient of -0.218 in column (3) implies deterring 1 firm in an industry with 5 incumbents. Overall, results suggests that entry of new firms after dereservation is negatively correlated with the trade credit terms in the industry.

The firms that managed to enter tend to match the trade credit terms with incumbent firms. Figure 5 plots the average of trade credit granted for incumbent firms and entrants after the dereservation. Two years after the dereservation, I find the difference between the average trade credit granted/sales for incumbent firms and entrants to be insignificant.

5 Robustness

In this section I consider various refinements of, and address potential concerns with, my baseline specification (Table 3, column (6)). I report the findings in Table 9. Panel A report results with *accounts receivables over sales* as dependent variable while panel B report results for *Log(Primary Product Price Index)*.

(i) *Potential Confounding Factors*

As discussed earlier in the data section, major changes in policies vis-à-vis foreign investment occurred in the early 1990s, and then stalled during the period of dereservation reform. Nataraj (2011) shows that tariffs were largely harmonized across industries by the late 1990s, so even though there were some reductions during the 2000s, the variation in tariff rates across product types had fallen dramatically by the start of 1997. I estimate column (6) of Table 3 for the sample beginning from 1997 and report result in column (1) of Table 9. The effect of removal of entry barrier on trade credit remains statistically and economically significant.

(ii) *Sub-period Analysis*

As most of the products in sample dereserved after 2003, column (2) reports the results for the sample beginning from year 2003. Results are robust to this sample selection

(iii) *Excluding Firms Manufacturing Chemical Products*

In column (3), I exclude firms manufacturing chemical products, an industry with many reserved/dereserved products, and find that results remain significant.

(iv) *State-year Fixed Effects*

Various state-specific economic conditions and other political factors may be correlated with special committee's decisions than product-level characteristics. I include state-specific year fixed to control for such unobservables (column (4)). The results show similar pattern with these modifications.

(v) *Alternative Definition of Incumbent Firms*

In column (5), I report the results after changing the definition of incumbent firms. I redefine the incumbent as a firm that manufactured a reserved product three years before it was dereserved as its primary product. The results are robust and quantitatively similar to the baseline results.

(vi) *Exclude Industries with Large Firms*

Large incumbent firms who are market leader, in order to protect their market power, may try to influence the decision of policy makers in the country. Therefore, the timing of dereservation may not be completely exogenous to firm policies in those industries/products. In column (6), I exclude, from the treatment group, the industries (at 5-digit) with large leader firms.

(vii) *Firm-level clusters*

Finally, in column (7), I correct standard errors for clusters at firm-level instead of NIC5-digit product level. Results are robust to these changes

Overall, I find that my basic result is robust i.e. with increased threat of entry, incumbent firms increase payment terms and cut prices.

6 Conclusion

This study examines how financial constraints affect incumbent supplier firms choice of extending more trade credit vis-à-vis offering price discounts when facing increased threat of entry by competitors (as distinct from actual entry), by using trade credit or price as a credible preemptive strategy to deter potential entrants. To test this hypothesis and its implications, I exploit, *plausibly* exogenous, staggered removal of product level entry barriers over time for Indian manufacturing firms. This reform triggered an increase in competition which reduced market concentration by 10% in affected industries in the subsequent 5-6 years. In a difference-in-differences setting, I find that an average incumbent firm extends 11% more trade credit and lowers price by 8.6% when the threat of entry increases. Interestingly, firms with deeper pockets offer longer terms on trade credit, while constrained firms rely on price discounts. Also, I find that financially constrained customers to take credit offers while unconstrained customers take price cuts. I further confirm my results using another policy change in the country that improved access to finance. This paper thus complements recent articles on trade credit and industry structure thereby extending the literature on trade credit for its role as a strategic tool to deter future entry.

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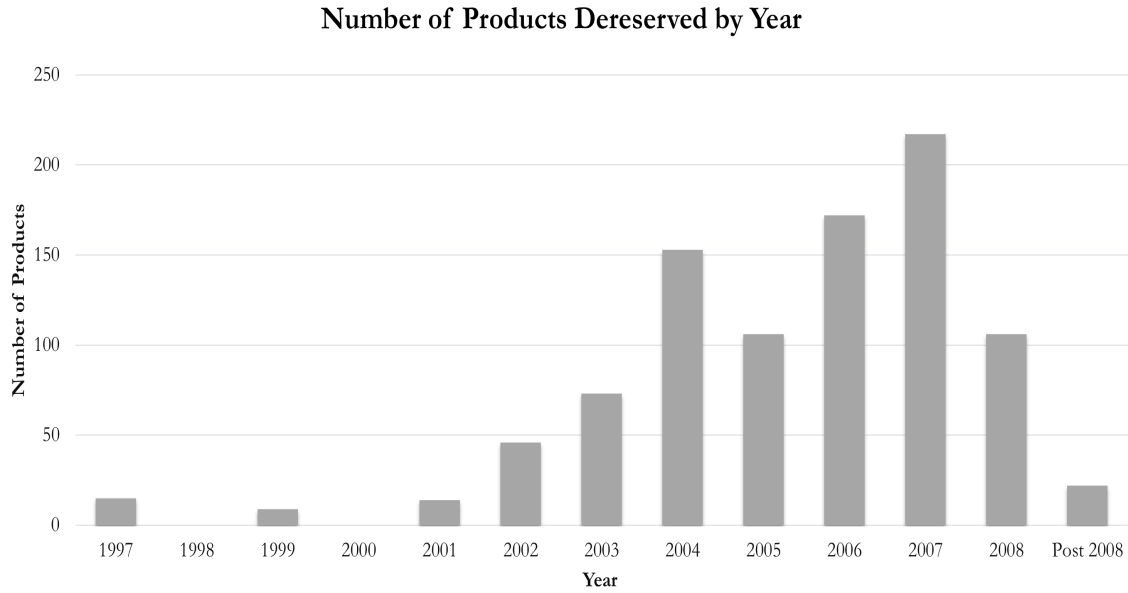
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Figure 1: Timing of Dereservation Reform

The figure below plots the number of products dereserved by Government of India over the years in different industries. Figure 1A plots the time-series from 1997 onwards. Figure 1B plots the number of products dereserved in each industry over the same period.

1A: Year-Wise



1B: Industry-Wise

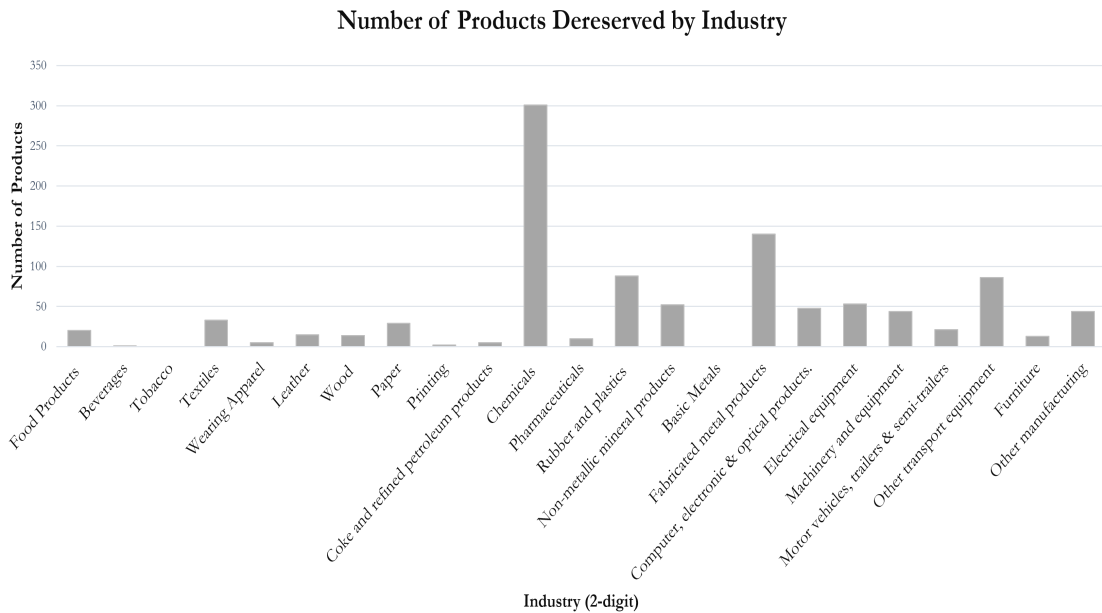
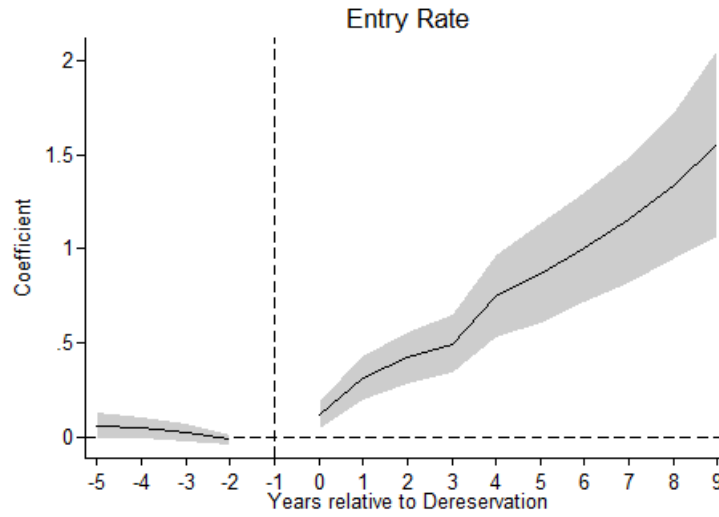


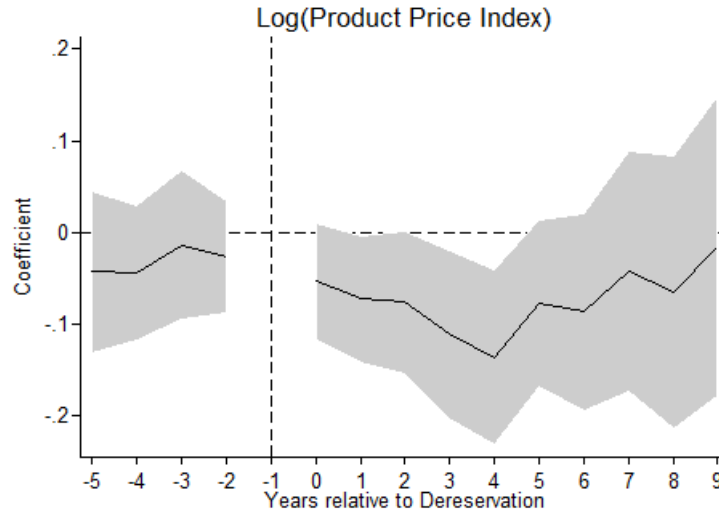
Figure 2: Effect of Dereservation on Industry Competition

The figures shows how dereservation is related to industry competition i.e. how entry rate, product prices and market concentration index (HHI) gets affected in reserved product category compared to a control group. The control group includes (a) pre-event observations of firms producing reserved products that were dereserved later and (b) firms whose products were never reserved/dereserved. I estimate a regression model similar to equation (1) with (a) Entry Rate (i.e number of new entrants scaled by incumbents), (b) Average Product Price Index (see section 3) and (c) Log(HHI) (measured at 5-digit NIC product-level) as dependent variable. The model also includes 5-digit product-level fixed effects and year fixed effects. Here, I plot the estimated coefficients for Pre-Dereservation ($k=-5$ to $k=-2$) to Dereservation ($k=0$ to $k=+9$) along with 1% confidence intervals around this difference.

2A: Entry Rate



2B: Product Price Index



2C: Market Concentration

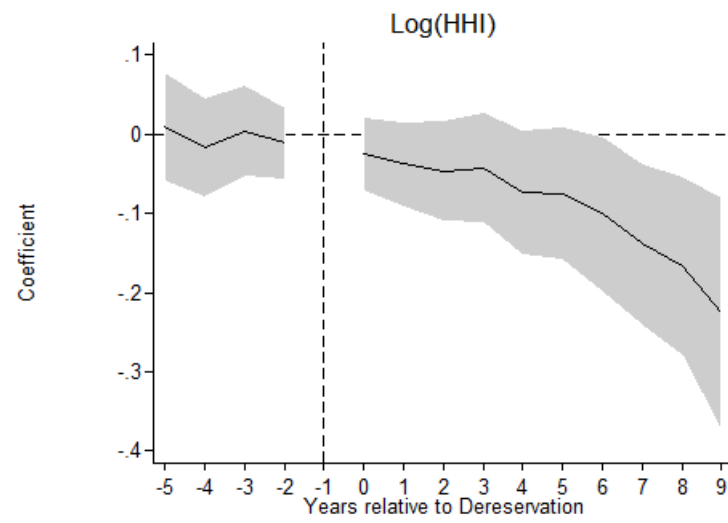


Figure 3: Effect of Dereservation on Trade Credit

The figure below plots the changes in mean trade credit granted (measured as accounts receivables scaled by sales) following dereservation. The figure shows how incumbent firms respond to dereservation within the reserved product category. I estimate equation (1) for dereserved products and plot the estimated coefficients for Pre-Dereservation ($k=-5$ to $k=-2$) to Dereservation ($k=0$ to $k=+9$) along with 1% confidence intervals around this difference.

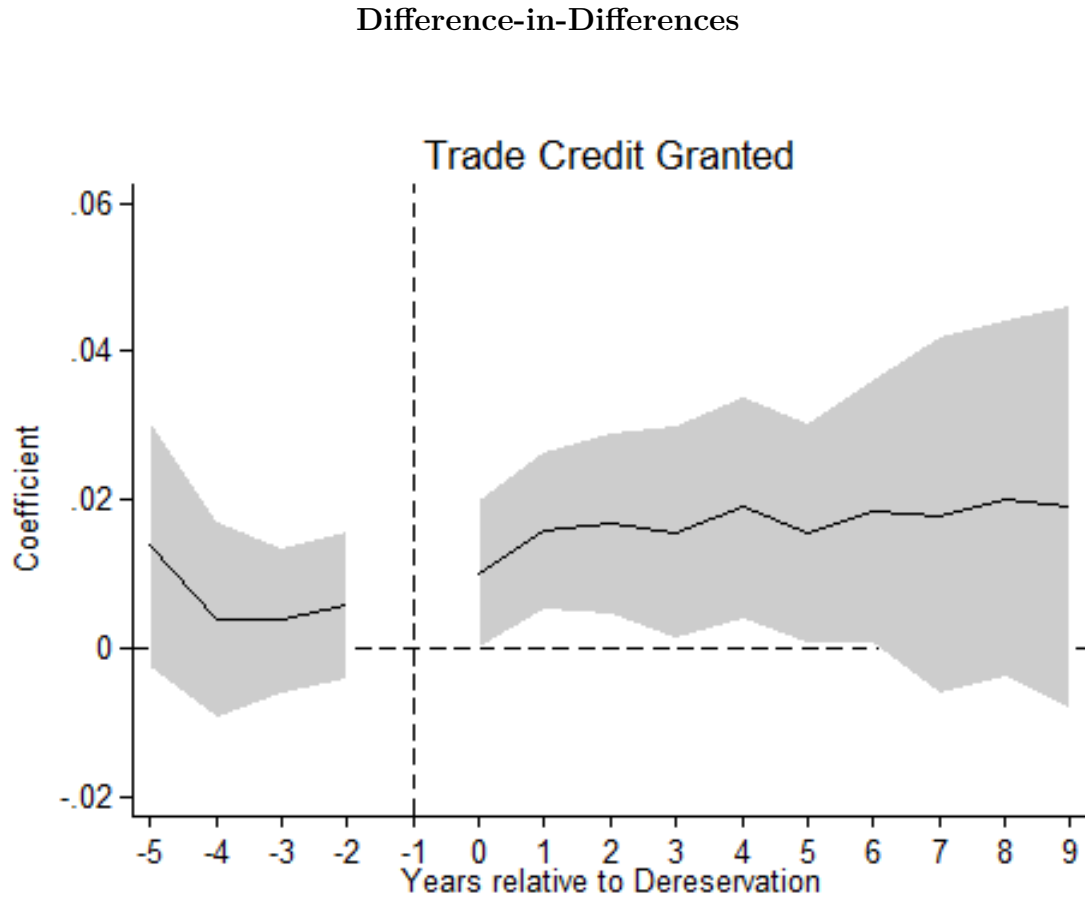


Figure 4: Non-Parametric Distribution of Placebo Dereservation Estimates

The figure below provides the result of a randomized removal of entry barriers of similar products, keeping the distribution of the event years unchanged. For each product dereserved (defined at 5-digit level) in the sample, I create a pseudo dereservation sample, where instead of the product that actually dereserved I assign it to one randomly selected similar product(i.e. at the same 4-digit level). In each of the pseudo dereservation samples I run my baseline regression as in Table 3, column (2) and save the relevant coefficients. Here, I plot the distribution of the coefficients. The black line embedded in the graph represents the regression coefficient obtained using the actual dereservation of products in the first year after the dereservation.

Placebo

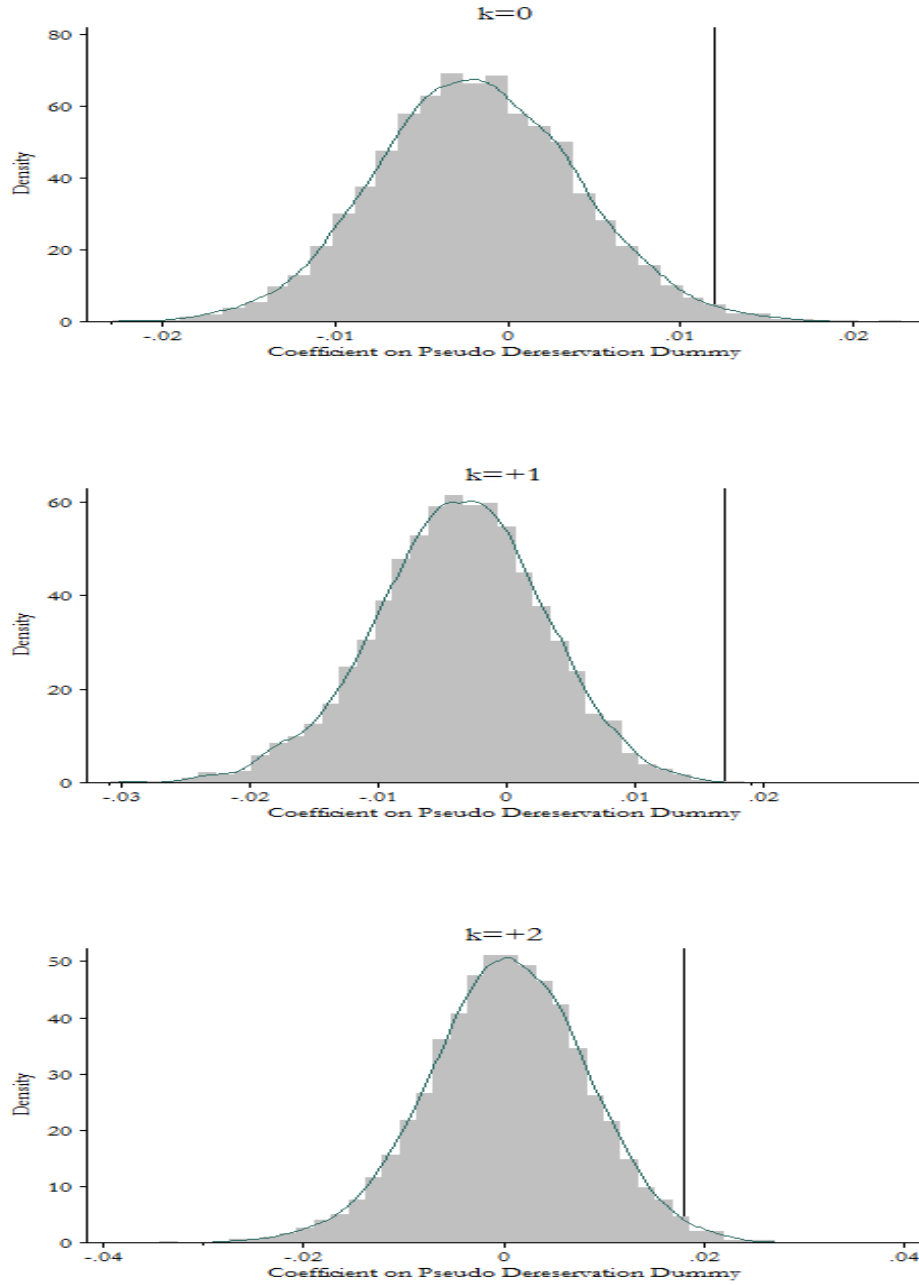


Figure 5: Ex-Post: Trade Credit Granted

The figure below plots the average trade credit granted (scaled by total sales) for both incumbents firms and new entrants after the dereservation.

Trade Credit Granted: Incumbents v/s Entrants

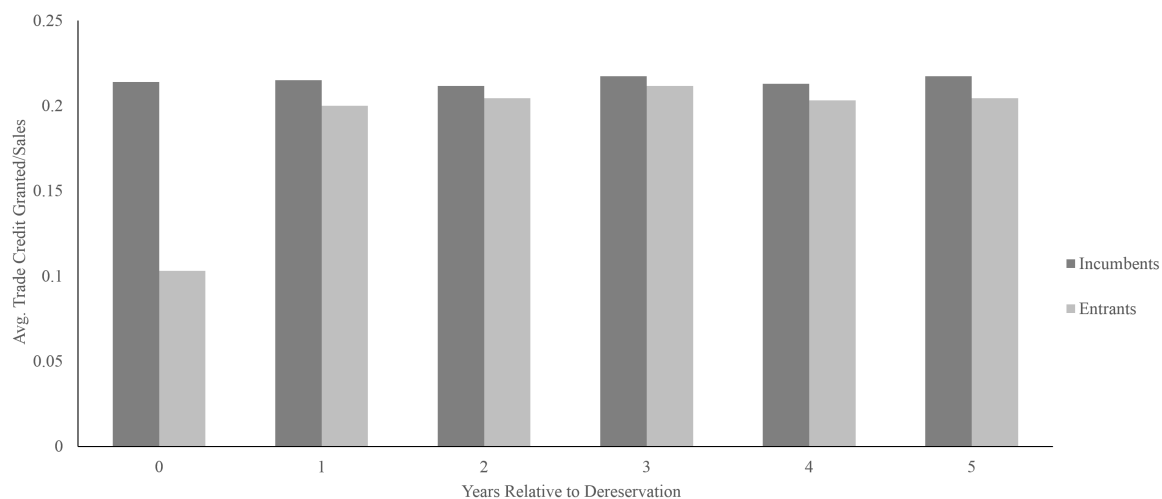


Table 1: Summary Statistics

This table reports descriptive statistics for my sample firms. I compile a firm-level panel data set based on the Prowess database, collected by the Centre for Monitoring the Indian Economy (CMIE). The database contains information primarily from the income statements and balance sheets of about 34,197 publicly listed/private companies, almost 10,724 of them in the manufacturing sector. Indian firms are required by the 1956 Companies Act to disclose product-level information on capacities, production, and sales in their annual reports. The CMIE compiles these detailed quantitative data and therefore enables me to track a firm's products over time. Furthermore, for each product manufactured by the firm, the data set provides the value of sales, quantity, and units. The Prowess is therefore particularly well suited for understanding how firms adjust their financial policies when their product lines are under threat of competition. The reservation policy was meant for only manufacturing firms in India, I keep only non-services, and non-government firms in my sample. All the financial variables are inflation adjusted using WPI and at 2004-2005 prices. The data also corrects for change in financial reporting year by adjusting values for number of months. To make sure that results are not effected by outliers, I winsorise all the ratios at 1% and drop observations with ratio of accounts receivables to sales greater than 1. Data period spans from beginning of the year 1990 to end of 2013.

Variables	Definition	Obs	Mean	Stdev.	Median
Trade Credit Granted	$\frac{\text{Accounts Receivables}}{\text{Sales}}$	32,603	0.19	0.15	0.16
Price of Output	Log of Primary Product Price Index	27,609	0.46	0.32	0.67
Trade Credit Received	$\frac{\text{Accounts Payables}}{\text{Assets}}$	40,188	0.18	0.16	0.13
Price of Input	Log of Primary Raw Material Price Index	32,454	0.24	0.25	0.61
Firm Size	Total Assets(in \$ Mn.)	32,603	53.2	200.6	11.1
Cash Holdings	$\frac{\text{Cash}}{\text{Total Assets}}$	32,603	0.06	0.09	0.03
Profitability	$\frac{\text{EBIT}}{\text{Total Assets}}$	32,603	0.12	0.10	0.12
Sales Growth	$\frac{\text{Sales}-\text{Lagged Sales}}{\text{Lagged Sales}}$	32,603	0.14	0.65	0.06

Table 2: Effect of Firm characteristics on Timing of Dereservation

This table reports results from Cox survival regressions that investigate the timing of dereservation of different products. The table looks at industry-level average firm characteristics. Firm characteristics are calculated as industry averages (at NIC5-digit) from 1990-1996 i.e. six years before the announcement of first product dereservation. All the specifications include industry fixed effects at 4-digit. Standard errors are corrected for heteroscedasticity and auto correlation, and clustered at NIC5-digit product level and reported in parentheses. *, **, and *** indicate significance at 10%, 5% and 1% respectively. All the variables are winsorized at 1%.

	<i>Time to Dereservation</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
Trade Credit Granted(Industry Avg.)	-1.240 (1.978)					3.683 (3.826)
Firm Size (Industry Avg.)		.251 (.312)				.387 (.286)
Profitability(Industry Avg.)			4.588 (4.875)			4.267 (3.685)
Cash Holdings(Industry Avg.)				.988 (2.296)		1.181 (3.319)
Sales Growth(Industry Avg.)					1.330 (1.142)	2.071 (1.307)
Industry Fixed Effects	4-digit	4-digit	4-digit	4-digit	4-digit	4-digit
Obs.	1065	1065	1065	1065	1065	1065
Log Pseudo likelihood	-241.64	-241.309	-241.303	-241.692	-240.477	-238.875
χ^2 statistic	16884.3	12924.58	1887.563	56474.67	1892.95	16223.03

Table 3: Base Line Results, Trade Credit Extended

This table presents the estimates from difference-in-differences regressions for *accounts receivables over sales*, proxy for payment terms (except column (4)), around the removal of entry barriers (dereservation of products). The estimates are based on the following regression equation:

$$y_{it} = \alpha_0 + \sum_{k=-2}^{-10} \beta_k \text{Pre-Dereservation}(k)_{it} + \sum_{k=0}^{10} \beta_k \text{Dereservation}(k)_{it} + \alpha_i + \alpha_t + \varepsilon_{it} \quad (1)$$

I define treatment as the elimination of small-scale reservation law on the incumbent firm's primary product in NIC-5 digit category. Firm's primary product is based on maximum sales revenue in the year preceding dereservation. *Pre-Dereservation*(k)_{*it*} (*Dereservation*(k)_{*it*}) is a dummy variable that takes a value one if it is k years before (after) the incumbent firm's primary reserved product has been dereserved, where incumbent firm is defined as a firm that ever made a reserved product before it was dereserved as its primary product. *Dereservation*(k)_{*it*} dummy identifies the group of firms under threat of entry (or exposure of firm to entry threat by using the proportion of total sales revenue from de-reserved product in pre-dereservation period.) The model is fully saturated with the year immediately before the dereservation as the excluded category. Therefore, the coefficients on *Pre-Dereservation*(k)_{*it*} (*Dereservation*(k)_{*it*}) compare the level of the dependent variable k years before (after) the dereservation to the year immediately before it's de-reservation. The inclusion of firm fixed effects, α_i , ensure that each indicator is estimated using only within firm variation in the dependent variable, and time dummies, α_t , control for country-level trends. In column (1) control group includes only pre-event observations of firms producing reserved products that were dereserved later (incumbents). In column (2) I include firms whose products were never reserved/dereserved and include an interaction between the year and incumbent dummies (incumbent $\times$ year fixed effects). Column (3), to estimate exposure of firm to entry threat, I replace dereservation dummies with the proportion of total sales revenue from de-reserved product in pre-dereservation period. Column (4) is like column (3) except with Log(1+Accounts Receivables) as dependent variable. Column (5) is like column (3) where I replace *incumbent* $\times$ *year* fixed effects with *incumbent* $\times$ *4-digit product* $\times$ *year* fixed effects. Column (6) reports results where to column(5), I further add *one year lagged* value of firm size (measured as log sales), cash holdings, profitability and sales growth as controls. Here I present estimated coefficients for Pre-Dereservation ($k=-3$ to $k=-2$) to Dereservation ($k=0$ to $k=+6$). Standard errors are corrected for heteroscedasticity and auto correlation, and clustered at NIC5-digit product level and reported in parentheses. *, **, and *** indicate significance at 10%, 5% and 1% respectively. All the variables are winsorized at 1%.

Trade Credit Granted (<i>Accounts Receivables over Sales</i>)						
	Only Treatment Firms	All Incumbents	Exposure to threat	Logged	Product Year FEs	Firm Controls
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=-3)	0.004 (0.005)	0.004 (0.005)	0.006 (0.006)	0.079 (0.060)	-0.002 (0.010)	0.004 (0.010)
Dereservation(k=-2)	0.006 (0.005)	0.007 (0.005)	0.009 (0.007)	0.083 (0.055)	0.014 (0.012)	0.016 (0.012)
Dereservation(k=0)	0.010** (0.005)	0.012** (0.005)	0.013** (0.006)	0.134** (0.054)	0.011 (0.010)	0.015 (0.011)
Dereservation(k=+1)	0.016*** (0.005)	0.017*** (0.006)	0.019*** (0.007)	0.129** (0.051)	0.024** (0.012)	0.026** (0.012)
Dereservation(k=+2)	0.017*** (0.006)	0.019*** (0.006)	0.019*** (0.007)	0.195*** (0.055)	0.020* (0.011)	0.022** (0.011)
Dereservation(k=+3)	0.016** (0.007)	0.017** (0.007)	0.017** (0.009)	0.208*** (0.063)	0.031** (0.015)	0.034** (0.015)
Dereservation(k=+4)	0.019** (0.008)	0.022*** (0.007)	0.020** (0.009)	0.219*** (0.067)	0.026* (0.016)	0.031** (0.015)
Dereservation(k=+5)	0.015** (0.008)	0.016** (0.007)	0.015* (0.009)	0.219*** (0.079)	0.029* (0.017)	0.034** (0.017)
Dereservation(k=+6)	0.018** (0.009)	0.020** (0.008)	0.017* (0.010)	0.204** (0.081)	0.024 (0.019)	0.029 (0.019)
Firm FEs	✓	✓	✓	✓	✓	✓
Year FEs	✓					
Incumbent × Year FEs		✓	✓	✓		
Incumbent × 4-digit					✓	✓
Product × Year FEs						
Firm Controls						✓
Obs.	13,989	31,918	31,918	31,918	31,161	31,161
R^2	0.634	0.652	0.652	0.874	0.701	0.709

Table 4: Price of Output

This table presents the estimates from difference-in-differences regressions for sample firms's Primary Product Price Index, around the removal of entry barriers (dereservation of products). I re-estimate equation (1) with $\log(\text{Primary Product Price Index})$. Column (1) report results with dereservation dummies and *incumbent* $\times$ *4-digit product* $\times$ *year* fixed effects. In column (2) I replace dereservation dummies with the proportion of total sales revenue from de-reserved product in pre-dereservation period. While in column (3) I add firm controls, as in table 3. In column (4)-(6), I adjust $\log(\text{Primary Product Price Index})$ for inflation using whole sale price index. Here I present estimated coefficients for Pre-Dereservation ($k=-3$ to $k=-2$) to Dereservation ($k=0$ to $k=+6$). Standard errors are corrected for heteroscedasticity and auto correlation, and clustered at NIC5-digit product level and reported in parentheses. *, **, and *** indicate significance at 10%, 5% and 1% respectively. All the variables are winsorized at 1%.

	Log(Primary Product Price Index)					
	Dummies	Exposure to threat	Firm Controls	Adjusted for Inflation		
				Dummies	Exposure to threat	Firm Controls
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation($k=-3$)	-0.003 (0.040)	-0.024 (0.044)	-0.029 (0.044)	-0.006 (0.041)	-0.026 (0.045)	-0.031 (0.045)
Dereservation($k=-2$)	-0.043 (0.033)	-0.055 (0.044)	-0.058 (0.044)	-0.040 (0.032)	-0.054 (0.043)	-0.056 (0.044)
Dereservation($k=0$)	-0.057 (0.040)	-0.069 (0.048)	-0.073 (0.048)	-0.060 (0.040)	-0.076 (0.047)	-0.080* (0.047)
Dereservation($k=+1$)	-0.094*** (0.031)	-0.101*** (0.039)	-0.105*** (0.039)	-0.092*** (0.032)	-0.102** (0.040)	-0.107*** (0.040)
Dereservation($k=+2$)	-0.068 (0.049)	-0.072 (0.058)	-0.077 (0.058)	-0.063 (0.049)	-0.074 (0.059)	-0.079 (0.060)
Dereservation($k=+3$)	-0.054 (0.056)	-0.057 (0.073)	-0.063 (0.074)	-0.040 (0.056)	-0.053 (0.074)	-0.060 (0.075)
Dereservation($k=+4$)	-0.072 (0.052)	-0.070 (0.075)	-0.078 (0.076)	-0.066 (0.049)	-0.075 (0.072)	-0.084 (0.072)
Dereservation($k=+5$)	-0.175** (0.071)	-0.201** (0.091)	-0.211** (0.091)	-0.162** (0.066)	-0.209** (0.087)	-0.219** (0.087)
Dereservation($k=+6$)	-0.094 (0.093)	-0.049 (0.123)	-0.059 (0.124)	-0.093 (0.086)	-0.072 (0.115)	-0.083 (0.116)
Firm FEs	✓	✓	✓	✓	✓	✓
Incumbent $\times$ 4-digit Product $\times$ Year FEs	✓	✓	✓	✓	✓	✓
Firm Controls			✓			✓
Obs.	23,506	23,506	23,506	23,506	23,506	23,506
R^2	0.74	0.74	0.741	0.788	0.788	0.788

Table 5: Access to Finance

This table reports the results of regressions estimating the differential response of incumbents firms to threat of entry based on their financial strength/access to finance. I use size-age (SA) index (Hadlock and Pierce (2010)) and a quasi-natural experiment i.e. establishment of Debt Recovery Tribunals (DRTs) in India (Gopalan, Mukherjee, and Singh (2016)), as proxy for financial strength/access to finance and report results in Panel A and B, respectively. For size-age index/DRT, I estimate column (6) of table 3 after replacing each *Dereservation(k)* (and *Pre-Dereservation(-k)*) with interaction terms *Dereservation(k) × Low SA/Late DRT* and *Dereservation(k) × High SA/Early DRT*, where *Low SA* is a dummy variable that identifies firms with below-median *Size-Age* Index in the year prior to dereservation of its primary product. While *Early DRT* identify the state of firm's registered office where DRTs were operational for at least five years in the year prior to dereservation of its primary product. In this specification, I also include a full set of interaction terms between different groups of financial strength and time fixed effects. This allows two groups of firms to have a differential time trend. Also, I control for industry specific time fixed effect at NIC4 level and add firm controls, for each group separately. In the column titled *Diff*, I test whether the coefficients estimated for different groups are significantly different. Note that in this test my sample is confined to treated firms that exist one year before the dereservation. Column (1)-(3) reports the results with *Accounts Receivables over Sales* as dependent variable, while column (4)-(6) reports the results with *Log(Primary Product Price Index)* adjusted for inflation, as dependent variable. Standard errors are corrected for heteroscedasticity and autocorrelation, and clustered at NIC5-digit product level and reported in parentheses. *, **, and *** indicate significance at 10%, 5% and 1% respectively.

Panel A: Size–Age Index

	<i>Trade Credit Granted</i>			<i>Log(Primary Product Price Index)</i>		
	Low SA	High SA	Diff	Low SA	High SA	Diff
	(1)	(2)	(2)-(1)	(4)	(5)	(4)-(5)
Dereservation(k=0)	-0.002 (0.015)	0.074** (0.035)	0.076** (0.034)	-0.136* (0.070)	-0.156 (0.138)	0.020 (0.136)
Dereservation(k=+1)	0.002 (0.015)	0.093*** (0.035)	0.091*** (0.033)	-0.159** (0.061)	-0.202 (0.139)	0.043 (0.150)
Dereservation(k=+2)	0.010 (0.018)	0.091** (0.035)	0.080** (0.034)	-0.195** (0.097)	-0.140 (0.150)	-0.055 (0.167)
Dereservation(k=+3)	0.023 (0.022)	0.087** (0.039)	0.064* (0.038)	-0.200 (0.141)	-0.103 (0.151)	-0.097 (0.168)
Dereservation(k=+4)	0.025 (0.025)	0.078** (0.038)	0.053 (0.036)	-0.241* (0.141)	-0.126 (0.145)	-0.115 (0.172)
Dereservation(k=+5)	0.018 (0.028)	0.084** (0.040)	0.065* (0.039)	-0.364*** (0.139)	-0.197 (0.157)	-0.167 (0.179)
Dereservation(k=+6)	0.013 (0.032)	0.080* (0.044)	0.066 (0.041)	-0.278* (0.165)	-0.187 (0.185)	-0.090 (0.191)
Firm FEs		✓			✓	
Incumbent × 4-digit		✓			✓	
Product × Year FEs						
Firm Controls		✓			✓	
Obs.		9,465			6,810	
R ²		0.91			0.84	

Panel B: Establishment of Debt Recovery Tribunals (DRTs)

	<i>Trade Credit Granted</i>			<i>Log(Primary Product Price Index)</i>		
	Late DRT	Early DRT	Diff (2)-(1)	Late DRT	Early DRT	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	-0.001 (0.017)	0.036 (0.028)	0.037 (0.029)	-0.435* (0.240)	-0.091 (0.060)	-0.345 (0.241)
Dereservation(k=+1)	0.016 (0.018)	0.049* (0.028)	0.033 (0.029)	-0.508** (0.241)	-0.114** (0.046)	-0.394 (0.247)
Dereservation(k=+2)	0.028 (0.020)	0.050* (0.029)	0.022 (0.029)	-0.484* (0.250)	-0.109 (0.074)	-0.375 (0.259)
Dereservation(k=+3)	0.050* (0.026)	0.045 (0.031)	-0.006* (0.032)	-0.574** (0.269)	-0.081 (0.112)	-0.493** (0.245)
Dereservation(k=+4)	0.059** (0.025)	0.035 (0.033)	-0.023 (0.032)	-0.579** (0.258)	-0.074 (0.101)	-0.505* (0.260)
Dereservation(k=+5)	0.051* (0.028)	0.035 (0.035)	-0.016 (0.034)	-0.694*** (0.250)	-0.135 (0.102)	-0.558** (0.252)
Dereservation(k=+6)	0.055 (0.035)	0.029 (0.037)	-0.026 (0.038)	-0.669** (0.285)	0.019 (0.123)	-0.688** (0.294)
Firm FEs		✓			✓	
Incumbent × 4-digit Product×Year FEs		✓			✓	
Firm Controls		✓			✓	
Obs.		9,448			6,793	
R^2		0.91			0.84	

Table 6: Financing Trade Credit

This table reports the results of regressions estimating the response of incumbents firms to threat of entry. I estimate column (6) of table 3 for different dependent variables. Column (1) -(4) report results with (a) Leverage, (b) Cash Holdings (scaled by total assets), (c) Trade Payable (scaled by assets) and (d) Debt Maturity (ratio of long-term debt to total debt), respectively. In column (5)-(7), I report the results of regressions estimating the response of incumbents firms to threat of entry based on firm's access to credit using a quasi-natural experiment i.e. establishment of Debt Recovery Tribunals (DRTs) in India (Gopalan, Mukherjee, and Singh (2016)). For DRT, I estimate column (6) of table 3 after replacing each *Dereservation(k)* (and *Pre-Dereservation(-k)*) with interaction terms *Dereservation(k) × Late DRT* and *Dereservation(k) × Early DRT*, where *Early DRT* identify the state of firm's registered office where DRTs were operational for at least five years in the year prior to dereservation of its primary product. In this specification, I also include a full set of interaction terms between different groups of financial strength and time fixed effects. This allows two groups of firms to have a differential time trend. Also, I control for industry specific time fixed effect at NIC4 level and add firm controls, for each group separately. In the column titled *Diff*, I test whether the coefficients estimated for different groups are significantly different. Note that in this test my sample is confined to treated firms that exist one year before the dereservation. Column (5)-(7) reports the results with *Debt Maturity* as dependent variable. Standard errors are corrected for heteroscedasticity and autocorrelation, and clustered at NIC5-digit product level and reported in parentheses. *, **, and *** indicate significance at 10%, 5% and 1% respectively.

	Leverage	Cash Holdings	Trade Payable	Debt Maturity	Debt Maturity		Diff
					Late DRT	Early DRT	(6)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dereservation(k=0)	-0.019 (0.030)	0.06 (0.05)	-0.02 (0.06)	0.013 (0.020)	-0.019 (0.023)	0.163*** (0.058)	0.182*** (0.060)
Dereservation(k=+1)	-0.012 (0.029)	0.04 (0.05)	0.04 (0.09)	0.012 (0.021)	-0.028 (0.033)	0.169** (0.067)	0.197*** (0.074)
Dereservation(k=+2)	-0.036 (0.027)	0.04 (0.06)	0.07 (0.09)	0.04 (0.024)	-0.01 (0.050)	0.117* (0.063)	0.119 (0.081)
Dereservation(k=+3)	-0.032 (0.046)	0.03 (0.07)	0.019* (0.011)	0.08 (0.020)	0.026 (0.051)	0.120* (0.065)	0.094 (0.085)
Dereservation(k=+4)	-0.024 (0.048)	0.04 (0.09)	0.020* (0.012)	-0.022 (0.026)	-0.014 (0.054)	0.111 (0.071)	0.125 (0.085)
Dereservation(k=+5)	-0.046 (0.048)	-0.04 (0.010)	-0.06 (0.014)	-0.030 (0.023)	-0.031 (0.053)	0.076 (0.075)	0.107 (0.088)
Dereservation(k=+6)	-0.036 (0.049)	0.06 (0.012)	0.05 (0.014)	-0.020 (0.026)	-0.01 (0.063)	0.123* (0.074)	0.124 (0.098)
Firm FEs	✓	✓	✓	✓		✓	
Incumbent × 4-digit	✓	✓	✓	✓		✓	
Product × Year FEs							
Firm Controls	✓	✓	✓	✓		✓	
Obs.	12,954	13,488	13,475	12,934		8,931	
R^2	0.709	0.635	0.73	0.583		0.76	

Table 7: Trade Credit Received and Price of Inputs

This table presents the estimates from difference-in-differences regressions for accounts payable/price of inputs around the removal of entry barriers (dereservation of products). The estimates are based on the regression equation (1), where I define treatment as the elimination of small-scale reservation law on the incumbent firm's primary *input* in NIC-5 digit category. Firm's primary *input* is based on maximum *raw material expenditure* in the year preceding dereservation. $Pre-Dereservation(k)_{it}$ ($Dereservation(k)_{it}$) is a dummy variable that takes a value one if it is k years before (after) the incumbent firm's primary reserved *input* has been dereserved, where incumbent firm is defined as a firm that ever used a reserved product before it was dereserved as its primary input. $Dereservation(k)_{it}$ dummy identifies the group of firms with more competition in the input market (or exposure of firm by using the proportion of total raw material expenditure from de-reserved product in pre-dereservation period). Panel A reports the results with *accounts payable/total assets* as dependent variable (except column (4) & (5)). In column (1) control group includes only pre-event observations of firms using reserved products that were dereserved later (incumbents). In column (2) I include firms whose primary inputs were never reserved/dereserved and include an interaction between the year and incumbent dummies (incumbent $\times$ year fixed effects). Column (3), to estimate exposure of firm to dereservation, I replace dereservation dummies with the proportion of total raw material expenditure from de-reserved product in pre-dereservation period. Column (4) is like column (3) except with $\text{Log}(1 + \text{Accounts Payable})$ as dependent variable. Column (5) is like column (3) except with *Accounts Payable scaled by Cost of Goods Sold* as dependent variable. Column (6) reports results where to column(3), I further add *one year lagged* value of firm size (measured as log sales), cash holdings, profitability and sales growth as controls. All regression are with *incumbent $\times$ 4-digit product $\times$ year* fixed effects (except column (1)). Here I present estimated coefficients for Pre-Dereservation ($k=-3$ to $k=-2$) to Dereservation ($k=0$ to $k=+6$). Panel B reports results from difference-in-differences regressions for sample firms's Primary Input Price Index, around the removal of entry barriers (dereservation of products). The specifications are similar to table 4 with $\text{Log}(\text{Primary Input Price Index})$ as dependent variable, where $Dereservation(k)_{it}$ is defined based on firm's input rather than output. Panel C and D of this table, like table 5, reports the results of regressions estimating the differential response of incumbents firms to dereservation of primary input based on their financial strength/access to finance. For size-age index/DRT, I estimate column (6) of table panel A after replacing each $Dereservation(k)$ (and $Pre-Dereservation(-k)$) with interaction terms $Dereservation(k) \times \text{Low SA/Late DRT}$ and $Dereservation(k) \times \text{High SA/Early DRT}$, where *Low SA* is a dummy variable that identifies firms with below-median *Size-Age Index* in the year prior to dereservation of its primary input. While *Early DRT* identify the state of firm's registered office where DRTs were operational for at least five years in the year prior to dereservation of its primary input. Column (1)-(3) reports the results with *Accounts Payable over Total Assets* as dependent variable, while column (4)-(6) reports the results with $\text{Log}(\text{Primary Input Price Index})$ adjusted for inflation, as dependent variable. Standard errors are corrected for heteroscedasticity and auto correlation, and clustered at NIC5-digit product level and reported in parentheses. *, **, and *** indicate significance at 10%, 5% and 1% respectively. All the variables are winsorized at 1%.

Panel A: Trade Credit Received

	Trade Credit Received (<i>Accounts Payable over Total Assets</i>)					
	Treated Firms	All Firms	Exposure	Logged	Scaled by COGS	With Controls
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=-3)	0.012 (0.009)	0.008 (0.009)	0.026** (0.011)	0.163 (0.101)	0.029 (0.019)	0.026** (0.011)
Dereservation(k=-2)	0.002 (0.009)	-0.001 (0.009)	0.003 (0.009)	-0.121 (0.080)	-0.007 (0.013)	0.003 (0.009)
Dereservation(k=0)	0.023** (0.009)	0.016* (0.009)	0.019* (0.010)	0.113 (0.085)	0.019 (0.013)	0.019* (0.010)
Dereservation(k=+1)	0.024*** (0.008)	0.016** (0.007)	0.024** (0.010)	0.195** (0.078)	0.029** (0.012)	0.025** (0.010)
Dereservation(k=+2)	0.024*** (0.009)	0.015** (0.006)	0.034*** (0.010)	0.234*** (0.086)	0.044*** (0.016)	0.032*** (0.010)
Dereservation(k=+3)	0.034*** (0.011)	0.022** (0.009)	0.029** (0.013)	0.331*** (0.088)	0.024 (0.016)	0.027** (0.012)
Dereservation(k=+4)	0.023* (0.013)	0.012 (0.009)	0.035* (0.019)	0.370*** (0.118)	0.049** (0.020)	0.033* (0.018)
Dereservation(k=+5)	0.006 (0.014)	0.005 (0.011)	0.004 (0.016)	-0.014 (0.121)	0.006 (0.015)	0.004 (0.016)
Dereservation(k=+6)	0.010 (0.016)	0.004 (0.011)	0.010 (0.018)	0.013 (0.149)	0.010 (0.016)	0.009 (0.018)
Firm FEs	✓	✓	✓	✓	✓	✓
Incumbent × 4-digit	✓	✓	✓	✓	✓	✓
Product × Year FEs						
Firm Controls						✓
Obs.	12,022	38,323	38,323	38,323	38,323	38,323
R^2	0.708	0.71	0.71	0.882	0.701	0.712

Panel B: Price of Inputs

	Log(Primary Input Price Index)					
	Dummies	Exposure	Firm Controls	Adjusted for Inflation		
				Dummies	Exposure	Firm Controls
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=-3)	0.064 (0.055)	-0.006 (0.067)	-0.007 (0.068)	0.066 (0.056)	-0.006 (0.067)	-0.007 (0.068)
Dereservation(k=-2)	0.010 (0.060)	-0.022 (0.096)	-0.022 (0.095)	0.007 (0.059)	-0.027 (0.094)	-0.027 (0.093)
Dereservation(k=0)	-0.019 (0.057)	-0.059 (0.069)	-0.061 (0.069)	-0.012 (0.056)	-0.046 (0.068)	-0.048 (0.068)
Dereservation(k=+1)	-0.129** (0.060)	-0.156** (0.078)	-0.161** (0.078)	-0.131** (0.056)	-0.151** (0.075)	-0.157** (0.076)
Dereservation(k=+2)	-0.114 (0.074)	-0.099 (0.087)	-0.105 (0.088)	-0.111 (0.069)	-0.087 (0.086)	-0.094 (0.086)
Dereservation(k=+3)	-0.047 (0.076)	-0.041 (0.090)	-0.048 (0.090)	-0.038 (0.070)	-0.021 (0.086)	-0.029 (0.087)
Dereservation(k=+4)	-0.113 (0.087)	-0.129 (0.113)	-0.137 (0.112)	-0.115 (0.081)	-0.122 (0.109)	-0.131 (0.109)
Dereservation(k=+5)	-0.175 (0.119)	-0.099 (0.170)	-0.104 (0.171)	-0.173 (0.108)	-0.096 (0.163)	-0.101 (0.164)
Dereservation(k=+6)	-0.206* (0.125)	-0.243 (0.164)	-0.247 (0.164)	-0.199* (0.113)	-0.228 (0.159)	-0.233 (0.159)
Firm FEs	✓	✓	✓	✓	✓	✓
Incumbent × 4-digit Product × Year FEs	✓	✓	✓	✓	✓	✓
Firm Controls			✓			✓
Obs.	31,458	31,458	31,458	31,458	31,458	31,458
R^2	0.743	0.743	0.743	0.767	0.767	0.767

Panel C: Size–Age Index

	<i>Trade Credit Received</i>			<i>Log(Primary Input Price Index)</i>		
	Low SA	High SA	Diff (2)-(1)	Low SA	High SA	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	0.017 (0.011)	0.003 (0.011)	0.015 (0.013)	-0.023 (0.078)	-0.083 (0.086)	-0.059 (0.081)
Dereservation(k=+1)	0.031** (0.014)	0.019 (0.015)	0.012 (0.015)	0.021 (0.081)	-0.107 (0.090)	-0.129 (0.103)
Dereservation(k=+2)	0.042** (0.017)	0.012 (0.015)	0.030 (0.022)	0.087 (0.085)	-0.073 (0.091)	-0.160 (0.103)
Dereservation(k=+3)	0.017 (0.015)	0.008 (0.017)	0.009 (0.023)	0.029 (0.095)	-0.103 (0.096)	-0.132 (0.127)
Dereservation(k=+4)	0.022 (0.027)	0.018 (0.023)	0.004 (0.021)	-0.023 (0.117)	-0.107 (0.119)	-0.084 (0.151)
Dereservation(k=+5)	-0.021 (0.019)	-0.019 (0.016)	-0.002 (0.024)	0.003 (0.133)	-0.082 (0.182)	-0.085 (0.176)
Dereservation(k=+6)	-0.004 (0.023)	-0.004 (0.019)	0.000 (0.023)	-0.143 (0.168)	-0.311* (0.175)	-0.168 (0.178)
Firm FEs		✓			✓	
Incumbent × 4-digit		✓			✓	
Product×Year FEs						
Firm Controls		✓			✓	
Obs.		6,649			4,780	
R^2		0.90			0.81	

Panel D: Establishment of Debt Recovery Tribunals (DRTs)

	<i>Trade Credit Received</i>			<i>Log(Primary Input Price Index)</i>		
	Late DRT	Early DRT	Diff (2)-(1)	Late DRT	Early DRT	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	0.132*** (0.038)	0.018 (0.011)	0.114*** (0.038)	0.012 (0.280)	-0.057 (0.079)	0.069 (0.296)
Dereservation(k=+1)	0.119*** (0.035)	0.035** (0.014)	0.084** (0.037)	0.128 (0.269)	-0.089 (0.079)	0.217 (0.277)
Dereservation(k=+2)	0.116*** (0.040)	0.025* (0.014)	0.091** (0.041)	0.131 (0.302)	-0.015 (0.083)	0.147 (0.324)
Dereservation(k=+3)	0.080* (0.044)	0.008 (0.015)	0.072 (0.045)	0.173 (0.309)	-0.048 (0.097)	0.220 (0.339)
Dereservation(k=+4)	0.085* (0.045)	0.016 (0.023)	0.068 (0.046)	0.207 (0.363)	-0.104 (0.099)	0.311 (0.382)
Dereservation(k=+5)	0.012 (0.041)	0.011 (0.015)	0.001 (0.044)	0.166 (0.442)	-0.017 (0.147)	0.182 (0.485)
Dereservation(k=+6)	-0.001 (0.043)	0.015 (0.019)	-0.016 (0.047)	0.190 (0.473)	-0.346* (0.194)	0.537 (0.494)
Firm FEs		✓			✓	
Incumbent × 4-digit		✓			✓	
Product×Year FEs						
Firm Controls		✓			✓	
Obs.		6,757			4,757	
R^2		0.90			0.81	

Table 8: Entry Deterrence

This table reports the results of regressions estimating the differential effect of trade credit on entry rates. Here, I estimate a entry rate as ratio of number of entrants to number of incumbents in five-digit industry for a given year and estimate the base-line specification for group of firms which have increased (decreased) trade credit in the year and one year after the dereservation. The regression also include five-digit industry fixed effects and year fixed effects. Standard errors are corrected for heteroscedasticity and autocorrelation, and clustered at NIC5-digit product level and reported in parentheses. *, **, and *** indicate significance at 10%, 5% and 1% respectively.

	Entry Rate		
	Δ Trade Credit Granted > 0	Δ Trade Credit Granted ≤ 0	Diff
	k=0,1 (1)	k=0,1 (2)	(1)-(2)
Dereservation(k=0)	.058 (.039)	.148*** (.057)	-.09 (.065)
Dereservation(k=+1)	.212*** (.055)	.407*** (.092)	-.196** (.093)
Dereservation(k=+2)	.315*** (.092)	.533*** (.093)	-.218* (.116)
Dereservation(k=+3)	.402*** (.095)	.646*** (.110)	-.244*** (.093)
Dereservation(k=+4)	0.850*** (0.160)	0.873*** (0.168)	-0.023 (0.136)
Dereservation(k=+5)	0.887*** (0.185)	1.087*** (0.207)	-0.200 (0.153)
Dereservation(k=+6)	1.049*** (0.211)	1.281*** (0.269)	-0.232 (0.146)
Industry Fixed Effects		5-digit	
Year Fixed Effects		Yes	
Obs.		1,900	
R^2		0.57	

Table 9: Robustness Checks

This table provides robustness of the results for the base-line specification i.e. column (6) of Table 3. Panel A report results for *Accounts Receivables over Sales* as dependent variable, while panel B report results for *Log(Primary Product Price Index)*. Major changes in policies vis-a-vis foreign investment occurred in the early 1990s, and then stalled during the period of dereservation reform(i.e. after 1997). Column (1) reports the results for the sample beginning from year 1997. Most of the products in sample dereserved after 2003, Column (2) reports the results for the sample beginning from year 2003. In column (3), I exclude firms manufacturing chemical products. Column (4), I include state-year fixed effects. In column (5), I redefine the incumbent as a firm that made a reserved product three years before it was de-reserved as its primary product. In column (6), I exclude industries (at 5-digit) with large leader firms. Standard errors are corrected for heteroscedasticity and auto correlation, and clustered at NIC5-digit product level and reported in parentheses. Finally, in column (7), I correct standard errors for clusters at firm-level. *, **, and *** indicate significance at 10%, 5% and 1% respectively.

Panel A: Trade Credit Granted

<i>Accounts Receivables over Sales</i>							
	After 1997	After 2003	Exclude chemical products	State-year FEs	Treatment definition	Exclude Industries with Large Firms	Standard Errors
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dereservation(k=-3)	0.002 (0.012)	0.001 (0.014)	0.00003 (0.012)	0.006 (0.011)	0.002 (0.012)	-0.0006 (0.020)	0.004 (0.012)
Dereservation(k=-2)	0.012 (0.012)	0.019 (0.012)	0.015 (0.015)	0.014 (0.014)	0.013 (0.014)	-0.009 (0.028)	0.016 (0.012)
Dereservation(k=0)	0.012 (0.011)	0.024** (0.009)	0.013 (0.012)	0.014 (0.011)	0.013 (0.012)	0.011 (0.020)	0.015 (0.011)
Dereservation(k=+1)	0.023** (0.012)	0.036*** (0.012)	0.023* (0.012)	0.027** (0.013)	0.020 (0.012)	0.043* (0.024)	0.026** (0.012)
Dereservation(k=+2)	0.019* (0.011)	0.033*** (0.012)	0.018 (0.013)	0.023** (0.011)	0.016 (0.011)	0.037** (0.017)	0.022* (0.013)
Dereservation(k=+3)	0.028* (0.015)	0.047*** (0.016)	0.024* (0.014)	0.035** (0.015)	0.026** (0.013)	0.057** (0.029)	0.034** (0.015)
Dereservation(k=+4)	0.026* (0.015)	0.043** (0.018)	0.023 (0.016)	0.030** (0.015)	0.014 (0.015)	0.049* (0.029)	0.031** (0.015)
Dereservation(k=+5)	0.029* (0.017)	0.051*** (0.019)	0.031* (0.018)	0.035* (0.018)	0.018 (0.016)	0.064** (0.028)	0.034* (0.018)
Dereservation(k=+6)	0.023 (0.020)	0.048** (0.022)	0.019 (0.018)	0.028 (0.020)	0.011 (0.018)	0.056 (0.039)	0.029 (0.019)
Firm FEs	✓	✓	✓	✓	✓	✓	✓
Incumbent × 4-digit Product × Year FEs	✓	✓	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓	✓	✓
Obs.	26,018	18,294	26,728	31,008	31,161	24,594	31,161
R^2	0.727	0.772	0.714	0.723	0.709	0.715	0.709

Panel B: Price of Output

	<i>Log(Primary Product Price Index)</i>						
	After 1997	After 2003	Exclude chemical products	State-year FEs	Treatment definition	Exclude Industries with Large Firms	Standard Errors
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dereservation(k=-3)	-0.041 (0.043)	-0.061 (0.060)	-0.071 (0.054)	-0.053 (0.045)	-0.038 (0.042)	-0.083 (0.062)	-0.051 (0.051)
Dereservation(k=-2)	-0.019 (0.041)	0.020 (0.045)	-0.071 (0.056)	-0.019 (0.045)	-0.018 (0.046)	-0.047 (0.062)	-0.032 (0.047)
Dereservation(k=0)	-0.071 (0.048)	-0.051 (0.055)	-0.143*** (0.052)	-0.091* (0.055)	-0.071 (0.051)	-0.112 (0.087)	-0.086 (0.057)
Dereservation(k=+1)	-0.111*** (0.040)	-0.106* (0.054)	-0.128*** (0.049)	-0.115** (0.046)	-0.078* (0.043)	-0.194*** (0.070)	-0.121** (0.056)
Dereservation(k=+2)	-0.087 (0.058)	-0.081 (0.078)	-0.153** (0.075)	-0.095 (0.065)	-0.051 (0.063)	-0.137* (0.074)	-0.105* (0.061)
Dereservation(k=+3)	-0.057 (0.080)	-0.058 (0.104)	-0.090 (0.107)	-0.059 (0.085)	-0.022 (0.076)	-0.177** (0.083)	-0.082 (0.064)
Dereservation(k=+4)	-0.085 (0.071)	-0.091 (0.103)	-0.094 (0.081)	-0.059 (0.082)	-0.038 (0.074)	-0.227** (0.112)	-0.099 (0.068)
Dereservation(k=+5)	-0.180** (0.088)	-0.170 (0.115)	-0.174* (0.104)	-0.153 (0.094)	-0.165* (0.084)	-0.373*** (0.122)	-0.206** (0.087)
Dereservation(k=+6)	-0.073 (0.117)	-0.097 (0.146)	0.021 (0.108)	-0.023 (0.127)	-0.023 (0.101)	-0.411*** (0.154)	-0.093 (0.095)
Firm FEs	✓	✓	✓	✓	✓	✓	✓
Incumbent × 4-digit Product × Year FEs	✓	✓	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓	✓	✓
Obs.	19,045	12,672	19,890	23,324	23,500	18,956	23,500
R^2	0.787	0.826	0.788	0.799	0.788	0.799	0.788

Appendix A

The reporting of products by Indian firms is not governed by any particular product classification. Although CMIE has developed an internal product classification that is based on the Harmonized System (HS) and National Industry Classification (NIC) schedules. There are about 3514 codes based on the NIC and the HS schedule. The agency has explicitly linked the product names reported by the firms to this classification. The names of products reported by the firm could differ in aggregation, or even in spelling (e.g., "Steel Rod" versus "Steel Rods"). Most of these issues were taken care by CMIE. But still there were some discrepancies like "Millstone, grindstone" versus "Millstone, grindstone, etc." which I cleaned up manually. CMIE has standardized approximately 51,320 product names to 3,714 possible CMIE product codes.

My final sample includes 2710 product codes out of the universe of 3714 product codes. The unused codes are products in the agriculture and services sectors, which, of course, are not produced by manufacturing firms. These 2710 product codes are mapped to 591 five-digit NIC product codes in 137 four-digit NIC industries across the 24 manufacturing sectors (two-digit NIC codes). As a comparison, the US data used by Bernard, Redding and Schott (2011) contain approximately 1,500 products, defined as five-digit Standard Industrial Classification (SIC) codes, across 455 four-digit SIC industries. Thus, my definition of a product is slightly more detailed than that of Bernard, Redding and Schott (2011).

I created a concordance between the NIC product codes and the list of reserved and de-reserved products. I matched products reserved to each firm based on 5-digit industry. In most cases, the match between NIC codes and SSI codes/product codes was exact that I was able to create the match based solely on the product descriptions. In other cases, I used the lengthy descriptions associated with the industry codes to help resolve many questionable concordances. I assumed that a product was matched to an NIC code if it was at least a partial match. Table A1 shows sample notification about list of items dereserved in 1997 while table A2 provides information on the number of products dereserved in a 2-digit NIC industry for the period 2003-2008. Next, table A3 shows the mapping of product codes in dereserved list with NIC codes. All the notifications about dereservation are available at <http://www.dcmsme.gov.in/publications/notiissforresderes.html>.

Examples of products within the Food Product sector (NIC 10) of this hierarchical mapping are listed in Table A4. The table reports two industries within the sector: Manufacture of Bakery Products, which contains 3 products (at NIC 5 digit level), and Manufacturing of Sugar, which contains 9 products. The product classification provides a concordance to the more familiar NIC industry codes used to classify economic activity in India. Each of the 2710 product codes can therefore be mapped to a five-, four-, three-, two-, or one-digit NIC code. Several features of the product data give me additional confidence in its quality despite the self-reported and non-standardized nature of the dataset. First, firms are required to report not just the names of the products, but also product-level details about installed capacity, production, sales quantity and value. The product-level data are available for 85 percent of the firms; this accounts for more than 90 percent of output and exports of the manufacturing firms in Prowess. More importantly, the product-level information and overall output are in separate modules of the Prowess database which enables me to cross check the consistency of the data. The data span the period from 1991 to 2013.

Table A1: Sample of Dereservation Notification

Here I provide sample notification about list of items dereserved in 1997. All the notifications about dereservation of products are available at <http://www.dcmsme.gov.in/publications/notiissforresderes.html>

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असाधारण
EXTRAORDINARY

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MINISTRY OF INDUSTRY
(Department of Industrial Policy and Promotion)

NOTIFICATION

New Delhi, the 3rd April, 1997

S. O. 298 (E).—In exercise of the powers conferred by sub-section (2A) of section 29B of the Industries (Development and Regulation) Act, 1951 (65 of 1951) and after considering the recommendations made by the Advisory Committee constituted under sub-section (2B) of the said section, the Central Government hereby direct that the following amendments shall be made in the notification of the Government of India in the Ministry of Industry, Department of Industrial Development No. S.O.477(E), dated the 25th July, 1991, namely :—

In the said notification, in Schedule-III under the heading—LIST OF ITEMS RESERVED FOR EXCLUSIVE MANUFACTURE IN SMALL SCLAE SECTOR, the following items and the entries relating thereto shall be omitted, namely :—

Sl. No. (As per gazette Notification)	Product Code	Name of the product
1	2	3
1	201801	Ice Cream
4	202901	Vinegar
5	204200	Rice Milling
6	204300	Dal Milling
8	20530101	Biscuits
15	21620101	Poultry feed except in pellet form
16C	224302	Synthetic syrups
54	280906	Corrugated paper & boards
571	363708	Hair driers—all types
629	374717	Hub caps—auto
635	374737	Ornamental fittings—auto
644	37478101	Spot lamps assembly—auto
645	37478201	Stop lamps assembly—auto
646	3747870	Tail lamp assembly—auto
647	3747804	Ash trays—Car fittings"

Excluding combination lamp assembly

Table A2: Timing of Dereservation Reform by Industry-Year

The table below provides the information on the number of products dereserved in a 2-digit NIC industry for the period 2003-2008.

2-digit	Industry/ Year	Number of Products Dereserved					
		2003	2004	2005	2006	2007	2008
10	Food Products	0	2	0	2	0	6
11	Beverages	0	0	0	0	0	0
12	Tobacco	0	0	0	0	0	0
13	Textiles	0	0	33	0	0	0
14	Wearing Apparel	2	0	0	0	1	0
15	Leather Products	7	0	0	0	0	0
16	Wood Products	0	5	0	0	8	0
17	Paper Products	3	8	0	0	0	17
18	Printing	0	0	0	0	0	1
19	Coke and refined petroleum products	0	0	2	1	0	1
20	Chemicals	57	93	0	50	2	41
21	Pharmaceuticals	0	0	0	7	1	1
22	Rubber and plastics	0	1	20	0	26	11
23	Non-metallic mineral products	0	3	0	3	39	2
24	Basic Metals	0	0	0	0	0	0
25	Fabricated metal products	0	24	0	45	33	5
26	Computer, electronic & optical products.	0	2	0	1	6	2
27	Electrical equipment	0	3	28	2	7	11
28	Machinery and equipment	0	5	3	10	13	2
29	Motor vehicles, trailers & semi-trailers	0	4	0	9	4	0
30	Other transport equipment	0	0	4	35	42	0
31	Furniture	0	1	0	6	2	2
32	Other manufacturing	1	2	1	1	33	4

Table A3: Sample of Product Matches

Following table provides the sample of matches of product codes in dereserved list with NIC codes and dereservation dates.

Item No.	Product Code	Product Name	NIC Activity Name	NIC5 Digit	Dereservation Date
1	201801	Ice Cream	Manufacture of ice-cream and kulfi etc.	10505	3 Apr, 1997
3	202501	Pickles & Chutneys	Manufacture of pickles, chutneys, murabbas etc.	10306	10 Apr, 2015
5	204200	Rice Milling	Rice milling	10612	3 Apr, 1997
6	204300	Dal Milling	Dal milling	10613	3 Apr, 1997
7	205101	Bread	Bread making	10711	10 Apr, 2015
242	312405	Chlorinated paraffin wax	Paraffin wax	19202	10 Oct, 2008

Table A4: Examples of Industries, Sectors and Products

NIC 1071 contains 3 products and NIC 1072 contains 9 products.

Group	Class	Sub-class	Description
	1071		Manufacture of bakery products This class excludes: - manufacture of farinaceous products (pastas), see 1074 - manufacture of potato snacks, see 1030 - heating up of bakery items for immediate consumption, see division 56
		10711	Manufacture of bread
		10712	Manufacture of biscuits, cakes, pastries, rusks etc.
		10719	Manufacture of other bakery products n.e.c.
	1072		Manufacture of sugar This class excludes: - manufacture of glucose, glucose syrup, maltose, see 1062
		10721	Manufacture or refining of sugar (sucrose) from sugarcane
		10722	Manufacture of 'gur' from sugarcane
		10723	Manufacture of 'gur' from other than sugarcane
		10724	Manufacture of 'khandsari' sugar from sugarcane
		10725	Manufacture of 'khandsari' suger from other than sugarcane
		10726	Manufacture of 'boora' and candy from sugarcane
		10727	Manufacture of 'boora' and candy from other than sugarcane
		10728	Manufacture of molasses
		10729	Manufacture of sugar from other sources (juice of palm, suger beet etc.)